SINGLE-VOLUME REVISED EDITION

FROM OPEN INTELLIGENCE TO PREDICTION

A practical guide to evidence analysis, Transformer forecasting, and accountable pandemic intelligence reports.

UNDERSTAND

BUILD

VALIDATE

FOUNDATIONS / OSINT DATA / TRANSFORMERS / FORECAST REPORTING

32 chapters | applied appendices | synthetic worked case

Educational reference / Not a live public-health forecast

EDITORIAL NOTE

A single book, three disciplines

The book you are holding consolidates three overlapping treatments of the same problem: the analyst's task of judging evidence, the engineer's task of building a reliable prediction service, and the researcher's task of proving that a forecast deserves attention. Material repeated across the earlier editions has been removed. The remaining chapters follow one continuous learning sequence from basic concepts to advanced Transformer systems.

01 / LEARN

For new analysts

Build a vocabulary for claims, observations, descriptive statistics, epidemiology and source reliability.

02 / BUILD

For engineers

Design as-of data pipelines, supervised forecasting datasets, attention models, and reproducible releases.

03 / TEST

For researchers and leaders

Evaluate calibration, compare baselines, examine uncertainty and connect forecasts to decisions.

Scope and use

All district case numbers and model outputs are fictional examples. Nothing here is a diagnosis, a real-time pandemic warning, a clinical directive or a claim of demonstrated model skill. A live system would require data permissions, public-health expertise, dataset-version controls, measured predictive performance and an authorized reviewer.

Data portals, public-health definitions and model libraries change. Verify release calendars and definitions before operational use. WHO notes that COVID-19 reporting counts can change with surveillance practices; CDC's FluSight evaluations show why performance should be inspected across forecast horizons and changing epidemiological phases.

Source updates: data.who.int/dashboards/covid19/data | cdc.gov/flu-forecasting

PREFACE

The forecast is not the model

At the start of an outbreak, the analyst rarely receives a tidy dataset. A hospital dashboard is revised overnight. A newspaper publishes a dramatic anecdote whose geography is imprecise. A policy announcement is posted after markets have already responded, and a wastewater series arrives with missing weeks. A machine learning model can process these observations; it cannot decide, on its own, which claims are trustworthy or which actions are legitimate.

That is why the forecasting problem begins with a question asked by a person: *What future outcome would change our decision?* The answer must name an outcome, location, unit, reporting period, information cut-off and forecast horizon. Those simple coordinates are more consequential than an impressive architecture diagram. Without them, even an accurate-looking curve may describe a quantity nobody needs.

Open-source intelligence broadens the available context. It can reveal policy changes, emerging operational pressure, shifts in testing and explanations for anomalous observations. But a published claim is not the same as a measured event. Several web articles can ultimately trace to the same source; a timestamp on a website is not necessarily the date of the event; a reported case series is not the true incidence of infection. We therefore move deliberately from source criticism and data contracts into forecasting mathematics.

Transformers enter only after that foundation. Attention can represent relationships across temporal positions, patches or variables. Temporal Fusion Transformer, PatchTST and iTransformer make different assumptions about what a token means and which covariates are available. None is universally best. This book treats them as competing experimental designs, always compared against transparent baseline forecasts and scored under historically realistic information constraints.

The final product is not a tensor or a notebook. It is an intelligence report a reviewer can audit: what was observed, what was forecast, how uncertain the prediction is, which assumptions are consequential, which alternatives were considered, and which decision may reasonably follow. The numeric forecast must be frozen before prose is drafted; generative language may assist presentation, but must not invent sources, probabilities, dates or causal explanations.

If one habit survives the more technical chapters, let it be this: every sentence in a prediction report should have a traceable role. Some sentences describe evidence, some communicate the model's predictive distribution, and some express human judgment. A professional report makes those boundaries visible.

HOW TO USE THIS VOLUME

One question, three lenses

The chapters should be read in order on a first pass. Afterward, the engineer can use the schemas and code as a prototype, the researcher can use the validation protocol as a release test, and the analyst can use the report template as a review checklist.

LEARN

What is the evidence?

Define observation, claim, target and horizon.

BUILD

Can we reproduce it?

Record data vintages, models and generated outputs.

TEST

Does it earn trust?

Run retrospective tests and review decisions.

A production prediction chain

1 **Source portfolio** — 2 **As-of evidence** — 3 **Models + baselines**

4 **Score + calibrate** — 5 **Signed forecast** — 6 **Reviewed report**

Release gates

Evidence gate: Provenance and known-availability time recorded; no privileged or future data entered.

Model gate: Rolling-origin comparison passes predeclared baseline, calibration and subgroup tests.

Publication gate: The forecast packet is immutable, provenance is linked and human review is documented.

GUIDE TO CHAPTERS

Contents

PARTS I-III FOUNDATIONS, DATA, AND FORECAST VALIDATION

PART I Reasoning Before Modeling

01 The question comes before the model 007
02 How information becomes intelligence 009
03 A practical language of numbers 011
04 Epidemiology without unnecessary jargon 013
05 Credibility, provenance and corroboration 015
06 Ethical boundaries and professional judgment 017

PART II From OSINT to Reliable Data

07 Signals are not all surveillance 019
08 Mapping a global OSINT source portfolio 021
09 Collecting evidence with provenance 023
10 Time series and the calendar problem 025
11 Non-time-series information is still predictive context 027
12 Turning open-source text into structured events 029
13 Feature engineering without self-deception 031

PART III Forecasting as a Testable Claim

14 Forecast targets, lead times and granularity 033
15 Simple baselines are formidable competitors 035
16 Disease mechanisms and changing regimes 037
17 A forecast is a distribution 039
18 Backtesting without time travel 041
19 What accuracy means to a decision-maker 043

GUIDE TO CHAPTERS

Contents (continued)

PARTS IV-V TRANSFORMER MODELS AND THE DECISION REPORT

PART IV Transformer Forecasting in Practice

20 From linear algebra to neural forecasting 045
21 Self-attention, step by step 047
22 Sequences, patches and future covariates 049
23 Temporal Fusion Transformer 051
24 PatchTST and long-context learning 053
25 iTransformer and architecture selection 055
26 Joining time series and unstructured intelligence 057
27 Training, tuning and constrained outputs 059

PART V From Model Output to Intelligence Report

28 Ensembles, explanation and dissent 061
29 A production OSINT-to-forecast architecture 063
30 Drift, incidents and human oversight 065
31 Writing the prediction intelligence report 067
32 A complete synthetic district case 069

Appendices & reference

Appendix A / Data contracts: a source becomes a forecast input 071
Appendix B / A minimal Transformer you can inspect 074
Appendix C / A decision-ready pandemic intelligence brief 079
Appendix D / The model comparison workbook 083
Appendix E / Glossary of analysis and forecasting terms 086
Appendix F / Advanced methods 092
References / Annotated sources and primary research 100
Closing note / The forecast is an accountable argument 103

PART I / REASONING BEFORE MODELING

CHAPTER 01 / FOUNDATION

The question comes before the model

Define what the prediction is for, not merely what data are available.

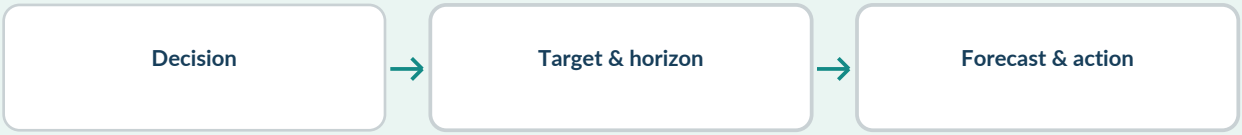
START WITH A REAL DECISION

A regional health planner asks whether wards will need extra capacity in four weeks. An analyst can answer with a chart of recent cases, a plausible scenario, or an actual forecast. Those are three different products. The conversation should begin with what the planner can change: staffing rosters, laboratory procurement, oxygen supply or public messaging. Only then can the analyst choose a target, horizon, geography and issue date. A beautifully trained model that predicts the wrong quantity is not a useful intelligence system.

The basic idea

A prediction is a statement about an observable future quantity. Start with five coordinates: the outcome (weekly new hospital admissions), the population and geography (one region), the time unit (epidemiological week), the lead time (one through four weeks ahead), and the publication cut-off (Friday at 12:00 local time). Specify whether the number is a raw count or a population rate. A forecast also needs a reference value against which improvement will be judged. This contract prevents teams from quietly switching targets when data become inconvenient.

CONCEPT MAP



Forecast origin	Horizon	Target
Last moment at which evidence is permitted into a forecast.	Distance from issue time to the future target.	Precisely defined quantity a model is scored against.

How the method works

Write the decision brief before opening a spreadsheet. Record the user, the decision deadline, the smallest useful change in forecast, and the cost of acting too early or too late. Define a threshold such as a weekly admission volume above which extra capacity may be considered, but ask clinicians and planners to set it. Freeze an explicit forecast origin. Separately define detection questions, such as whether an unfamiliar syndrome is appearing in reports. Detection locates possible events; forecasting estimates future measured outcomes. Mixing them without labeling the distinction produces false confidence.

The expert perspective

Expert systems distinguish forecasts, scenarios and counterfactuals. A forecast is conditional on information available at issue time; a scenario imposes hypothetical changes such as reduced testing or increased contact rates; a counterfactual asks what might have happened under an alternative intervention. A Transformer trained only on historic observations does not automatically know the effect of a future policy. If users want a what-if statement, supply assumptions and either a defensible causal model or clear scenario language. Measure the value of a forecast by decisions it supports, not architecture prestige.

WATCH OUT	A target without a dated definition invites accidental leakage and misleading comparisons.

Worked practice

Consider an invented district reporting 202 influenza-like hospital admissions last week. The planner needs a four-week outlook, not a one-year trend chart. The target sheet states: weekly admissions reported by hospitals, district A, issue date 8 October 2026, horizons 1-4 weeks, provisional data through the latest eligible week. It also states a capacity review level of 250 admissions per week. Every later table, model input and report headline should carry these definitions unchanged.

Apply the lesson

01	Interview the person who will act on the forecast and record their decision deadline.
02	Write a one-sentence target definition containing units, jurisdiction, date and horizon.
03	Choose a baseline and a no-release rule if the required data are unavailable.

IN THE REPORT	As of the stated issue date, we forecast weekly admissions for weeks 1-4 ahead, with uncertainty intervals and a stated capacity threshold.

CHAPTER 02 / FOUNDATION

How information becomes intelligence

Move from observation to evidence, inference and judgment.

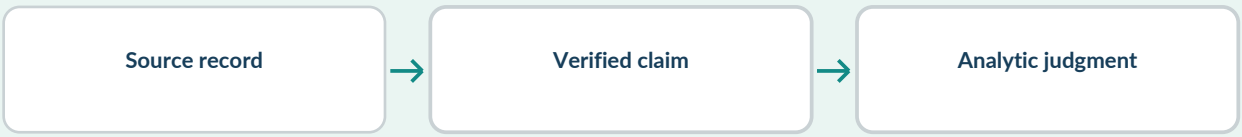
START WITH A REAL DECISION

A newspaper reports that a hospital is busy; a government table shows admissions; a social post claims that a new disease is spreading. All three may be relevant, but they do not have the same evidential status. The analyst's work is not to combine the words into one alarming sentence. It is to establish who said what, which claim can be tested, what is independently corroborated, and what remains unknown. Intelligence becomes valuable when another analyst can reproduce the path from source to conclusion.

The basic idea

A source is the origin of information. An observation is something measured or directly recorded. A claim is a proposition about the world. An inference is a conclusion drawn from evidence; a judgment acknowledges residual uncertainty. These categories should remain distinct in notes and final reports. Headlines often imply causes where the underlying reporting supports only coincidence. The simplest professional habit is to ask, for each sentence, "What would make this statement false?" If no possible observation could disconfirm it, it may be commentary rather than analysis.

CONCEPT MAP



Primary source
Original publisher or recorder of an observation.

Triangulation
Comparison across genuinely independent evidence streams.

Analytic judgment
Assessment that goes beyond directly observed facts.

How the method works

Use a claim ledger. Give every relevant assertion an identifier, link it to its original publisher and timestamp, record the claimed event location and event date, and assign a verification status. Separate primary records from later summaries that quote the same record. A small evidence table can carry claim, support, contradiction, confidence, provenance and next action. Analysts should conduct an alternative-hypothesis check before publication: data error, routine seasonality, a local outbreak or changed reporting practices might all explain an apparent surge. This discipline complements models rather than competing with them.

The expert perspective

Confidence in a factual claim is not the same as the probability of a forecast event. A report might have high confidence that a hospital changed its reporting definition, yet only moderate confidence in the numerical trajectory after the change. Avoid multiplying subjective confidence scores as though they were calibrated probabilities. Instead document which assumptions are verified, which are estimates, and what observations would change the analysis. Formal structured analytic techniques help expose unstated preferences, but they do not turn weak evidence into strong evidence.

WATCH OUT	Ten copied accounts are still one underlying observation.

Worked practice

Imagine three articles mention "40 new cases" in the same city. The articles trace back to one local authority statement. Counting three articles as three independent confirmations exaggerates the evidence. After deduplication the analyst records one attributed report, checks the date of the primary notice, and seeks a separate laboratory or admission series. The forecasting feature should represent one event and its attributes, not three correlated press mentions.

Apply the lesson

01	Create a ledger with a distinct row for a source, claim and underlying event.
02	Write one plausible alternative explanation for the strongest signal.
03	Mark each conclusion as observation, inference, judgment or forecast.

IN THE REPORT	Independent surveillance indicators are broadly consistent with rising activity, but causal explanations remain uncertain.

CHAPTER 03 / FOUNDATION

A practical language of numbers

Learn rates, distributions, denominators and uncertainty.

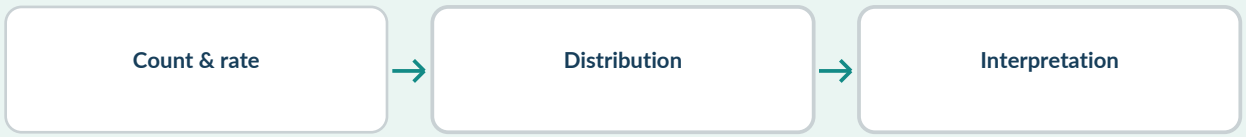
START WITH A REAL DECISION

Suppose district X reports twice as many admissions as district Y. That sounds decisive until someone reveals that district X has four times the population. An analyst comfortable with basic quantitative reasoning does not need sophisticated mathematics to catch the error. They need to know what was counted, what was eligible to be counted, and which denominator makes a comparison fair. Forecasting builds on this literacy: models amplify both useful signal and poorly defined numbers.

The basic idea

Distinguish counts from rates. A weekly admission count is useful for capacity planning, while admissions per 100,000 residents help compare districts. A percentage needs a defined numerator and denominator; a median describes a typical value differently from a mean when the distribution is skewed. Percent change can be unstable near zero. Correlation measures co-movement, not an intervention effect. An uncertainty interval describes a range of possible future outcomes under stated modeling assumptions; it is not a promise that nature respects the bounds.

CONCEPT MAP



Denominator
Population or tested group relative to which an event is counted.

Quantile
Value below which a specified fraction of a distribution falls.

Prediction interval
Range estimated to contain a future realization with stated coverage.

How the method works

A quick descriptive analysis starts with a table of dates, values, denominators and missing flags. Plot the raw series before smoothing, then check weekly seasonality, sudden revisions, missing periods and implausible jumps. Compare a current value with a relevant seasonal baseline, not merely last week's count. Write the calculation directly below the chart: rate = count / population x 100,000. When explaining distributions, show median and selected quantiles alongside the average. These elementary checks frequently reveal more than a first deep-learning run.

The expert perspective

Expert reviews examine selection bias and measurement error. A positive-test rate can rise because infection spreads, because tests target higher-risk patients, or both. A reported rate per resident can be distorted by cross-border care-seeking. Confidence intervals, credible intervals and prediction intervals arise from different inferential objects and should not be used interchangeably. Predictive intervals should be judged on future held-out outcomes, especially under regime shifts. Good communication uses probability statements tied to defined events rather than vague adjectives such as "likely" without a stated meaning.

WATCH OUT	A precise decimal attached to an uncertain denominator is not precision.

Worked practice

In a hypothetical region, admissions grow from 80 to 100; that is an increase of 20 admissions or 25%. If the region has 200,000 people, the latter week equals 50 admissions per 100,000 people. Another region with 120 admissions among 600,000 people has a lower rate of 20 per 100,000 despite a higher count. The correct comparison depends on whether the question concerns relative health burden or total ward beds required.

Apply the lesson

01	Compute a raw count, a population-standardized rate and a percentage change.
02	Plot an unsmoothed series and explain one spike in plain language.
03	Identify one denominator or sampling change that would reverse an initial conclusion.

IN THE REPORT	The raw count indicates greater absolute workload; the standardized rate indicates a different relative burden.

CHAPTER 04 / FOUNDATION

Epidemiology without unnecessary jargon

Understand transmission, observation and clinical severity.

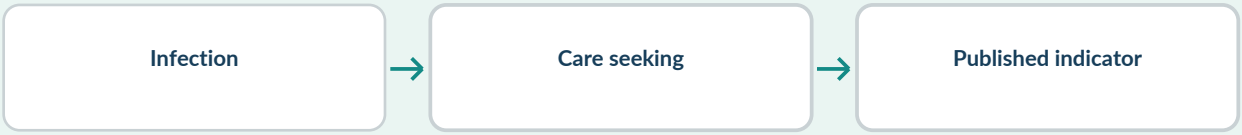
START WITH A REAL DECISION

A disease curve is not a direct photograph of infections. Infection happens first. Symptoms may follow, then testing, medical care, recording and publication. Each stage is filtered by behavior, clinical pathways and reporting systems. A model predicting published case counts therefore forecasts a combination of disease dynamics and observation dynamics. The distinction matters most when public messaging, access to tests or hospital reporting rules change quickly.

The basic idea

Basic concepts include incidence (new events in a defined period), prevalence (existing cases at a time), generation interval (time between infection of linked cases), and reproduction number (average secondary infections under specific conditions). Epidemic growth may accelerate, plateau or decay; those shapes can be affected by immunity, variants, behavior, vaccination, seasonality and interventions. Hospitalization is a lagged and selected outcome of infections. For planning, admissions may be more directly useful than estimated infection totals, but they still depend on clinical practice and surveillance completeness.

CONCEPT MAP



Incidence
Number of new events during a stated period.

Reporting lag
Delay from underlying event to data publication.

SEIR
Compartmental model separating susceptible, exposed, infectious and removed groups.

How the method works

Draw a causal pathway from exposure to hospitalization and a separate observation pathway from hospital event to published record. Place each available dataset on one of these paths. When wastewater viral signals lead clinical admissions, the lag must be estimated in training data, not imposed as a universal constant. Compare multiple outcomes: emergency-department visits, positive tests, laboratory detections and hospital admissions. Disagreement can be informative if it reflects a change in severity or measurement rather than model noise.

The expert perspective

Advanced analysis distinguishes renewal models, compartmental models such as SEIR, and purely statistical forecasters. A mechanistic model encodes assumptions about how transmission unfolds; a Transformer learns predictive patterns from observed sequences. Neither automatically estimates causal intervention effects. Under a novel pathogen, historically learned associations may fail sharply. Maintain a separate outbreak intelligence process capable of escalating unusual signals before historical patterns become statistically reliable. Collaboration with epidemiologists is part of model design, not merely final approval.

WATCH OUT	Reported cases are an observation process, not a complete census of infections.

Worked practice

Suppose reported cases fall while hospital admissions rise. It is tempting to declare that severity has increased. Alternative explanations include reduced testing, delayed hospitalization, changing age distribution or revisions. A careful report describes the divergence and names evidence needed to distinguish hypotheses. The forecast system may retain admissions as its operational target while tracking cases as one uncertain covariate.

Apply the lesson

01	Define incidence and prevalence using an example, and identify the time unit.
02	Sketch the disease-to-reporting pathway for one target.
03	List two reasons a leading indicator might stop leading.

IN THE REPORT	Admissions remain elevated despite lower reported cases; competing explanations include reporting changes and population mix.

SELECTED SOURCE / [03] World Health Organization

CHAPTER 05 / FOUNDATION

Credibility, provenance and corroboration

Evaluate open-source material without treating popularity as proof.

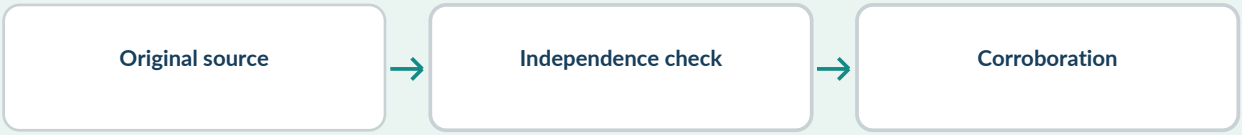
START WITH A REAL DECISION

Open-source intelligence is often abundant precisely when the situation is least clear. The same unverified claim may travel through social media, radio, news sites and an automated aggregator within minutes. An analyst needs to reconstruct that route before treating the information as multiple signals. Source reliability is not a permanent label: an institution may publish excellent weekly data but produce informal commentary that requires separate verification.

The basic idea

Evaluate authority for a particular claim, not prestige in general. Ask whether the source observed the event directly, whether definitions are public, how frequently the series is updated, and whether past corrections are visible. Provenance includes the original URL or document, author or agency, observation date, publication date and your retrieval time. A clear record of uncertainty is a mark of careful analysis. "Unverified" should remain a live status until corroboration arrives, rather than quietly turning into "fact" through repetition.

CONCEPT MAP



Provenance Recorded origin and transformation history of information.	**Syndication** Redistribution of one source through other publishers.	**Coverage** People, facilities, geography or events represented by a dataset.

How the method works

Operational corroboration has several stages. First locate the earliest accessible original record. Second separate syndication from independent collection. Third compare with adjacent indicators; for example, reported clinic congestion with measured admissions, if comparable. Fourth flag contradictions for human review. Source ranking can assist triage, but model features should preserve missingness, revisions and source identity. A defensible source catalog states coverage, latency, known bias and use restrictions for every feed. This is especially important when global datasets vary by country.

The expert perspective

At expert level, consider dependence graphs: two apparently independent data portals may share the same reporting ministry. A knowledge graph can represent entities, claims, citations and evidence lineage, but it should not automatically resolve contradictions. Statistical weighting by historical source quality may help, yet the quality labels themselves can be biased toward well-covered regions. Do not let a large-language-model summarizer infer unstated certainty from authoritative-looking prose. A human analyst should be able to click from each narrative claim back to its underlying document.

WATCH OUT	Audience engagement is not evidence of accuracy.

Worked practice

A viral post claims "hospitals overwhelmed nationwide" and cites a chart with no date. A local hospital dashboard reports a moderate rise in admissions. The appropriate report is not a nationwide assertion. It records an unverified claim, limits the observed rise to the measured facilities and investigates geographic representativeness. The model's input should not encode an unverified national surge as a fact.

Apply the lesson

01	Find the primary publication behind two apparently separate accounts.
02	Grade a source on coverage, latency, definitions and revision policy.
03	Trace one final report sentence back to the original archived record.

IN THE REPORT	The widely circulated claim is not independently confirmed by available hospital surveillance data.

SELECTED SOURCE / [01] World Health Organization

CHAPTER 06 / FOUNDATION

Ethical boundaries and professional judgment

Make the system useful without exposing people or manufacturing certainty.

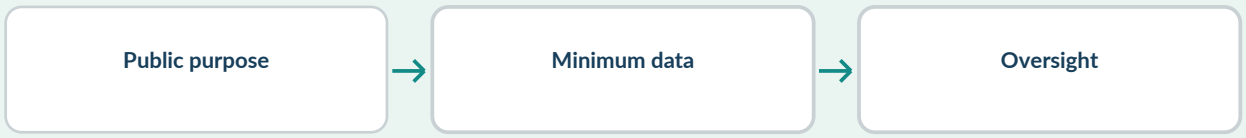
START WITH A REAL DECISION

Publicly accessible does not mean ethically consequence-free. A social post about symptoms may expose a person's health status even when technically visible on the internet. A local outbreak map can stigmatize a neighborhood if generated from a handful of reports. The forecasting mission should be defined at a population level and collect only information necessary for that mission. The team's ability to gather data is not by itself a justification for gathering it.

The basic idea

Ethical OSINT begins with purpose limitation, minimization, lawful access and proportionality. Prefer aggregated public-health series over individual records. Respect platform access rules and database licenses; document sensitive-source restrictions before ingestion. Do not publish private health details, contact information or potentially identifying clusters. Explain when a dataset may systematically miss rural areas, under-resourced communities or people outside formal healthcare. The duty to communicate uncertainty is also ethical: false precision can distort public-health decisions.

CONCEPT MAP



Data minimization
Collecting no more data than the declared purpose requires.

Purpose limitation
Restricting reuse to justified and authorized aims.

Feedback effect
Forecast-triggered action that changes later observations.

How the method works

Create a data-use review for every feed. Record its public-health purpose, lawful basis or permission status, level of aggregation, retention window, allowed users and publication constraints. Apply access controls to raw documents, even when public, if they include unnecessary personal details. Publish model outputs at a geographic and temporal resolution appropriate to the evidence. Make a human reviewer responsible for sensitive escalations and for any decision to suppress potentially harmful detail. See WHO surveillance ethics guidance and NIST AI risk-management resources in the bibliography.

The expert perspective

Advanced governance examines feedback effects. A forecast can change behavior and therefore change the outcome it forecasts: hospitals may expand capacity, people may seek testing sooner, and policymakers may issue interventions. This is not ordinary forecasting error alone; it changes the system being modeled. Document anticipated feedback loops, avoid portraying policy-sensitive scenarios as neutral inevitabilities, and evaluate differential error across regions. Governance also requires an incident path for harmful or erroneous reports, with an archive of prior versions and corrected statements.

WATCH OUT	A technically legal scrape can still be a poor public-health choice.

Worked practice

Imagine a map that highlights a tiny town because three symptom posts were scraped during one evening. Even without names, readers may infer household identities. A responsible team aggregates the signal to a safer level, treats the count as unverified and checks whether the report has legitimate decision value. The headline cannot be "outbreak confirmed" on the basis of these posts.

Apply the lesson

01	Identify what personal data the system can avoid collecting altogether.
02	Document retention and access rules for the riskiest feed.
03	Write a correction procedure for a forecast published with wrong source data.

IN THE REPORT	We report aggregated signals only; individual-level claims have not been used to estimate risk.

SELECTED SOURCE / [06] World Health Organization · [14] NIST

PART II / FROM OSINT TO RELIABLE DATA

CHAPTER 07 / FOUNDATION

Signals are not all surveillance

Differentiate indicators, events, rumors and operational measurements.

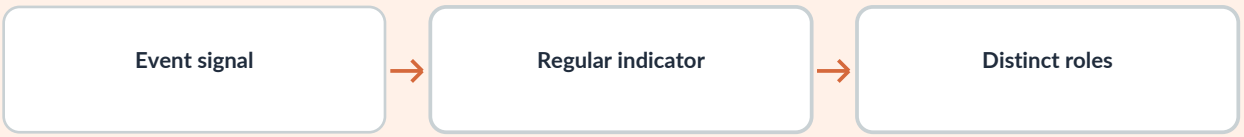
WHY THIS COMPONENT EXISTS

The first sign of an unusual cluster may be an emergency physician's remark in local news. The best measure of its scale may arrive later from laboratory or hospital systems. These sources answer different questions, and neither should be forced into the other's role. Event-based surveillance can alert analysts to what might be emerging. Indicator-based surveillance measures defined variables repeatedly. A forecasting system benefits from both when their roles are kept distinct.

Core contract

Indicators are observations defined over a regular time base: weekly admissions, test positivity or syndromic visits. Events are discrete reports: a new testing policy, a clinic closure, a suspected cluster or a newly identified strain. A static descriptor, such as age structure, changes infrequently and characterizes a jurisdiction. Each requires different validation. Indicator values need units and denominators; events need entity resolution and verification; static records need an effective date and a source. Text-only data are not automatically numerical predictors.

CONCEPT MAP



Indicator-based surveillance
Routine measurement of predefined health indicators.

Event-based surveillance
Detection and assessment of discrete potentially significant reports.

Availability time
Earliest moment an observation could enter a forecast.

Implementation contract

Create a source-to-use matrix with four columns: source, observation unit, earliest availability, and modeling role. The primary target should usually be a comparatively stable and decision-relevant observed series. Leading indicators can enter as lagged covariates after their historical predictive value is verified. News and policy events should first be extracted into auditable records with effective dates and uncertainty. Keep "signal detected" as a separate task from "predicted admissions" so unusual news does not become an unverified quantitative surge.

The expert perspective

Expert analysts test for look-ahead bias created by retrospective event coding. A news article published two weeks after an outbreak cannot be treated as a signal available before the outbreak, even if its text refers to an earlier event date. Likewise a final revised hospitalization series cannot be used as training inputs at past forecast origins without recreating historical vintages. Multi-source fusion should increase evidence quality, not defeat temporal integrity. Independent signals are especially valuable for detecting measurement regime changes.

WATCH OUT	The date an event happened is not always the date it became knowable.

Engineering exercise

An article published on 18 March says infections began "in early March." The article's event date is approximate, but its availability to a forecaster is 18 March. Using a 5 March timestamp as a model input on 10 March silently imports future information. The system should store both dates, as well as retrieval timestamp and the confidence of event extraction.

Apply the lesson

01	Label five sample records as indicator, event, static descriptor or unverified claim.
02	State whether each record predicts an outcome or merely describes the context.
03	Test whether a reported event was available at the earlier forecast origin.

IN THE REPORT	An event-based signal warrants review, while available indicator series do not yet establish its scale.

SELECTED SOURCE / [01] World Health Organization · [03] World Health Organization

CHAPTER 08 / INTERMEDIATE

Mapping a global OSINT source portfolio

Choose complementary feeds rather than collecting every possible feed.

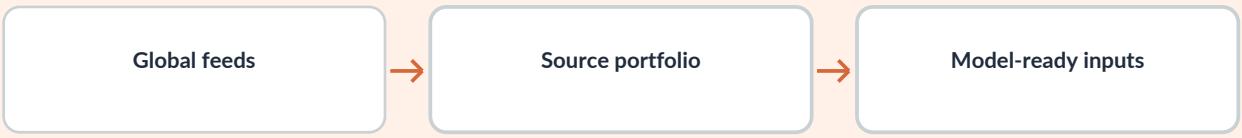
WHY THIS COMPONENT EXISTS

A global collection plan can become a shopping list: WHO, CDC, ministries, journals, news agencies, mobility reports, climate data, wastewater networks and dozens of social platforms. More sources do not necessarily improve forecasts. Each feed brings a different geography, refresh cycle, license, meaning of an observation and history of revisions. The useful question is whether the source answers a missing part of the target decision.

Core contract

Separate the portfolio into surveillance targets, leading health indicators, contextual drivers, event intelligence and reference metadata. Examples include WHO's public-health intelligence work, national surveillance portals, CDC respiratory-hospitalization datasets, demographic statistics and publicly documented policy announcements. Consult each provider's latest definitions and export availability before engineering an integration. A dashboard showing a trend is not necessarily a stable machine-readable API, and a well-known historic COVID-19 dataset may no longer receive daily updates.

CONCEPT MAP



Source portfolio Selected set of feeds serving defined analytical roles.	**Coverage bias** Systematic gaps in represented populations or places.	**Schema drift** Change in field names, meanings or allowed values over time.

Implementation contract

Use a scorecard: target relevance, geographical coverage, update interval, reporting delay, historical depth, reproducibility, license, and likely independence from other feeds. Prioritize two reliable series over ten thin ones. For international comparisons, maintain a crosswalk for case definitions, jurisdiction boundaries, reporting week conventions and population denominators. When an official feed breaks, quarantine it and inform users instead of substituting unrelated news volume. Preserve original units and provide a translation layer rather than forcing all feeds into one generic "risk" score.

The expert perspective

An expert portfolio includes leading and lagging relationships that may vary by variant, season and jurisdiction. Wastewater concentrations, where available, can carry different measurement and normalization issues from hospital admission counts. Climate variables may be useful for seasonal patterns but require careful geography and publication handling. Historical source quality should be evaluated separately by region and period; global average reliability hides weak local coverage. Formal provider agreements or public access policies may be necessary for production access.

WATCH OUT	Never promise that a historically famous data feed is still maintained without checking.

Engineering exercise

Suppose a district has a weekly admissions target, daily emergency-visit percentages and scattered news reports. The first two are candidates for quantitative modeling after alignment; the news may support anomaly review but should not be counted as an independent weekly hospitalization measure. A second news aggregator that republishes the same stories adds little source diversity.

Apply the lesson

01	Build a shortlist of one target, two leading indicators and two contextual feeds.
02	Document the observation unit, access method and last verified update for each.
03	Write a reason to exclude at least one tempting but weak source.

IN THE REPORT	The current source portfolio supports weekly admissions modeling; news feeds remain contextual and review-only.

SELECTED SOURCE / [01] World Health Organization · [04] US Centers for Disease Control and Prevention · [15] WHO

CHAPTER 09 / INTERMEDIATE

Collecting evidence with provenance

Design ingestion for repeatability and later audit.

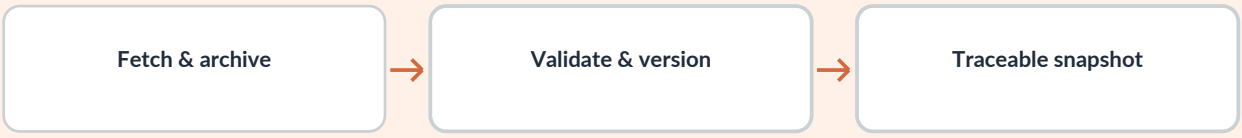
WHY THIS COMPONENT EXISTS

An analyst downloads the same spreadsheet on Monday and Thursday and sees different values for a week in the past. Which value did yesterday's published report use? If the system only stores the latest file, the answer may be unknowable. A production pipeline treats collection as an evidence-preservation step rather than a temporary precursor to feature engineering.

Core contract

A raw record should carry a source identifier, original URI, retrieved timestamp, publication timestamp if available, nominal observation period, checksum, file format and ingestion job version. Raw copies are immutable; cleaned tables are derived artifacts. Observability includes collection failures and partial loads, not merely successful API calls. Every transformation should be reproducible against archived inputs. Time zones and reporting week conventions must be explicit, particularly for global feeds.

CONCEPT MAP



Immutable raw zone
Preserved source files that are not overwritten by transformations.

Data vintage
Snapshot of what values were published at a given time.

Idempotence
Repeated ingestion does not duplicate or corrupt records.

Implementation contract

Build connector adapters that perform rate-limited retrieval, schema validation, type checking, checksums and idempotent storage. Snapshot incoming files before normalization. Store a manifest linking source version, file hash, row count and validation result. Create a quarantine lane for unexpected formats, excessive missingness or definition changes. A human source steward decides whether a change is harmless. Capture metadata needed for attribution and access compliance. The pipeline should be able to reconstruct the exact input table used for any issued forecast.

The expert perspective

Experts often need a bitemporal design: the time a value purports to describe and the time that value became known to the system. This matters for historical validation, incident review and explanation of changed forecasts. Event records need separate publication and event timestamps; indicator records often need observation end, release date and revision time. In complex systems a data contract, versioned table format and lineage graph can make these rules enforceable. Software reliability is not a substitute for source validity, but poor reliability can erase it.

WATCH OUT	Overwriting a revised figure destroys evidence about what was previously known.

Engineering exercise

If weekly admissions for week 34 were initially 180, revised to 194 and later finalized at 198, all three releases should remain traceable. A forecast issued after the first release must be evaluated using what it actually saw as input, even if final target truth is later set to 198 for scoring. The difference should be explained in an audit record.

Apply the lesson

01	Define a raw-source manifest with retrieval time and content hash.
02	Specify two data-quality conditions that block ingestion.
03	Reconstruct the input used by a past forecast from its version identifiers.

IN THE REPORT	This forecast used source vintages archived at the stated cut-off, with known revisions recorded separately.

SELECTED SOURCE / [03] World Health Organization

CHAPTER 10 / INTERMEDIATE

Time series and the calendar problem

Make timestamps, delays and seasonal patterns first-class data.

WHY THIS COMPONENT EXISTS

One agency reports calendar weeks ending Sunday; another reports epidemiological weeks ending Saturday. One daily feed closes at midnight UTC; another uses local time. Joining these by a casually formatted date can create an artificial lead or lag that looks predictive in model training. Before asking a Transformer to find temporal patterns, the analyst must decide what each timestamp means.

Core contract

A time series is a sequence indexed by a meaningful and consistent time reference. It can be daily, weekly or irregular, but the modeling frequency must be declared. Important time fields include event date, observation period end, public release, retrieval and revision. Lags describe how one series is shifted relative to another. A trailing average summarizes recent history; it is not a future value. Missing periods, zero values and late reports mean different things and should not be collapsed into one number.

CONCEPT MAP



Time series
Measurements indexed by ordered observation periods.

Nowcast
Estimate of a recent incompletely observed quantity.

Temporal leakage
Future information entering a past model input.

Implementation contract

Create a canonical time index for each target and map external feeds to it using documented rules. Aggregate daily values to weeks only when the measure is additive; averaging rates and summing counts are different operations. Carry data completeness flags and the number of contributing days. Build features with strictly past observations, for example a four-week trailing mean ending at the forecast cut-off. Estimate reporting delay curves where source revisions are material. Plot revisions by age since initial release before deciding whether to nowcast incomplete recent weeks.

The expert perspective

Temporal integrity extends to model evaluation. A backtest using the final latest historical table gives the model observations that were unavailable at many past issue dates. The result is an optimistic retrospective exercise, not a genuine simulated forecasting record. Consider revision-aware nowcasting and origin-specific feature extraction. For a multi-jurisdiction panel, handle differing holidays, seasonal cycles and time zones rather than imposing an artificial synchronous update. Temporal availability masks are as important as missing-value masks.

WATCH OUT	Correctly sorted dates can still conceal future information.

Engineering exercise

An agency publishes a final count on Wednesday for the preceding Monday-Sunday week. A briefing issued Monday morning cannot use that future Wednesday release, even though its observation period ended the day before. A feature builder must query records by release time, not only the event-week key.

Apply the lesson

01	Identify observation, publication and retrieval timestamps in a sample feed.
02	Compute a trailing mean using only values available by a specified cut-off.
03	Describe how a late revision would alter a backtest.

IN THE REPORT	Recent observations are provisional and were adjusted only using information released by the issue cut-off.

SELECTED SOURCE / [07] Hyndman & Athanasopoulos

CHAPTER 11 / INTERMEDIATE

Non-time-series information is still predictive context

Use stable, slowly changing and categorical data carefully.

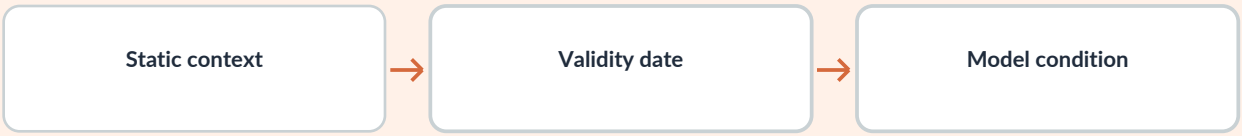
WHY THIS COMPONENT EXISTS

Some of the most important attributes of a region are not updated every week. Age structure, hospital-bed inventory, urbanization and typical care-access patterns may explain why two similar infection waves produce different admission counts. These features can improve transfer between locations, but they must be treated as contextual measurements rather than observations of current disease activity.

Core contract

Non-time-series inputs include static attributes such as population size, categories such as region type, geographic relationships, and slowly changing attributes such as vaccination coverage or capacity estimates. A value is only "static" relative to the forecast horizon, not for all time. Every contextual record needs a source, jurisdiction mapping, definition, and effective or survey date. Unknown attributes should be represented as missing rather than assigned the average of neighboring regions without disclosure. Text metadata, source trust and policy descriptions may also fit this class after validation.

CONCEPT MAP



Static covariate
Attribute treated as constant across the forecast window.

Effective date
Date when a contextual value becomes applicable.

Ecological fallacy
Mistaking group-level relationships for individual-level effects.

Implementation contract

Link contextual features through stable geographic identifiers, not free-form city names. Normalize counts to meaningful scales where appropriate, but do not hide the operational unit of the target. For time-varying covariates, build valid-from and valid-to intervals and join only records in force at the forecast origin. Apply train-only encoding and scaling. In a pooled model, static embeddings can supply regional context while a temporal encoder learns each location's changes. Test models with and without these attributes to see whether they truly improve forecasts.

The expert perspective

A subtle failure occurs when the final annual value of a context variable is used for all earlier months in retrospective training. That leaks an updated measurement into earlier origins. Another problem is ecological inference: a regional age proportion describes the population, not a particular patient. Transformer variable-selection scores are not causal evidence that a demographic characteristic drives risk. Fairness reviews should examine whether context features improve accuracy uniformly or merely help better-covered districts.

WATCH OUT	A value copied backward from a later survey can corrupt a historical forecast.

Engineering exercise

A rural district and urban district have identical recent admissions but different bed capacity and age mix. The model may benefit from these contextual differences when predicting counts; the report must still explain that capacity planning and disease risk are different quantities. Static features influence conditional predictions but should not be translated into demographic blame.

Apply the lesson

01	Separate truly fixed features from slowly changing observations.
02	Create a validity-date join for a regional capacity attribute.
03	Run a feature ablation that removes all static covariates.

IN THE REPORT	Regional context informs the estimate but does not explain individual outcomes or establish causality.

CHAPTER 12 / INTERMEDIATE

Turning open-source text into structured events

Use NLP and LLMs for extraction, not as an oracle.

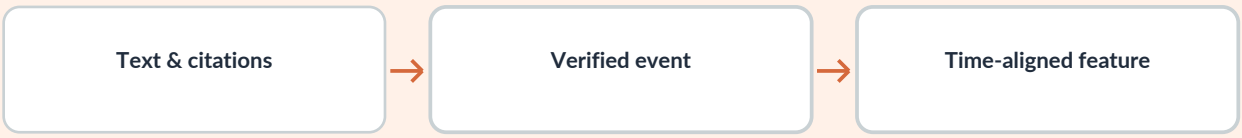
WHY THIS COMPONENT EXISTS

A news article may mention a temporary school closure, a local respiratory cluster and three quoted institutions in the same paragraph. A numerical forecasting model cannot directly use this document without an explicit representation. A language model can help extract candidate entities and events, yet it can also hallucinate dates or merge distinct incidents. The quality of the final feature depends on verification and time alignment, not the sophistication of the text model.

Core contract

A basic event schema might contain event_id, event_type, location_id, publication_time, event_time, source_id, verification_status, severity and confidence in extraction. Duplicate documents should resolve to one underlying event when evidence supports it. Separate what the source directly asserts from the extraction system's interpretation. Embeddings provide compressed semantic representations useful for retrieval and clustering; an embedding is not a clinical measurement. Most text-derived signals are sparse, selectively reported and sensitive to media attention.

CONCEPT MAP



Entity resolution
Matching mentions to the same real-world person, place or event.

Embedding
Vector encoding of text or other input.

RAG
Retrieval-augmented generation grounded in selected documents.

Implementation contract

Start with bounded extraction tasks: classify whether a document concerns public health, identify known event types, extract quoted dates and locations, then attach source spans. Require the extraction system to produce citations back to input text. Evaluate precision and recall on a manually annotated multilingual sample, including negation and rumor language. Use deterministic rules for known fields where they outperform more flexible generation. Aggregate verified events to a jurisdiction-week only after deduplication and publication-time controls. Retain explicit "no signal" versus "no coverage" states.

The expert perspective

Advanced fusion can encode verified event sequences or text embeddings alongside indicator streams, but source coverage creates strong selection bias. Media volume rises when public concern rises, which may correlate with the target without adding genuine leading information. Test against a simple article-count control, remove syndication clusters, and validate improvement by geography and time. Retrieval-augmented generation is appropriate for citing source passages in a report; it should not invent forecast numbers or replace statistical scoring.

WATCH OUT	A text model can create a plausible-looking event that the document never reported.

Engineering exercise

A classifier labels a headline "new outbreak" even though the article says authorities found no evidence of one. A human-verified annotation records "outbreak denied," preserving the negation. Treating the headline as an outbreak event would pollute both the feature table and the analyst narrative.

Apply the lesson

01	Extract two events and attach exact source snippets that justify the fields.
02	Deduplicate documents that describe a single underlying notice.
03	Test whether news features help more than a simple source-volume baseline.

IN THE REPORT	Verified policy and health events add contextual information; unverified media claims are excluded from numerical features.

CHAPTER 13 / INTERMEDIATE

Feature engineering without self-deception

Build modeling tables that reproduce a real forecast.

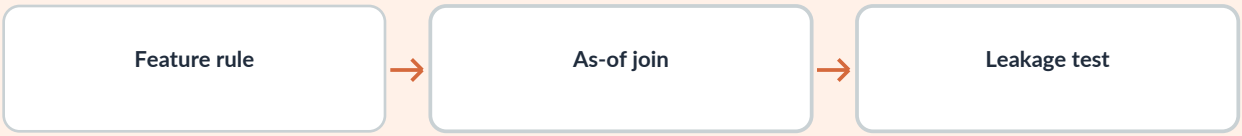
WHY THIS COMPONENT EXISTS

The forecasting table is where elegant analysis can fail quietly. A week labeled "2026-W40" may include latest revised values, a lagged article count, a static population figure and a forward-filled laboratory measurement. If the team cannot explain when each cell was available, it has not yet built a valid model input. A feature is not simply a column; it is a column with a definition, time rule and provenance.

Core contract

Common features include lagged target values, trailing means, seasonal terms, holidays, observed leading indicators, missingness flags, and static contextual variables. Known-future covariates, such as a calendar holiday, differ from unknown future observations, such as next week's positivity rate. Never pass unknown future values into a model during validation just because they exist in the finalized dataset. Scale training data using parameters estimated only on training periods. Encode missingness deliberately because an absent record can indicate either no events or a broken reporting process.

CONCEPT MAP



Feature registry
Contract describing how every model input is constructed.

Known-future covariate
Future input truly known when issuing the prediction.

Ablation
Controlled removal of a component to assess its contribution.

Implementation contract

Define a feature registry storing name, formula, unit, source, time alignment, missing-value rule and earliest availability. Build feature rows using an `as_of` query corresponding to each training origin. Store normalizer parameters with the model artifact. Unit tests should catch target leakage, duplicate keys, shifts caused by reporting calendars, and propagation of invalid values. Compare candidate features with interpretable ablations; removing a feature that has no measured value can improve robustness and maintainability. Use simple baseline feature sets before building a large panel.

The expert perspective

Transformers are sensitive to the semantics of feature arrangement. A positional encoding implies order; a missingness mask distinguishes padded steps from truly observed zeros. If some jurisdictions have short histories, training samples require consistent padding and mask logic. A feature known to be reported late should enter the model as the latest publicly available value, accompanied by age-since-release information. Consider robustness under alternate imputation policies and sudden source outages. More variables raise the risk of spurious associations without enough independent seasons to learn them.

WATCH OUT	Unknown future covariates must not be supplied as if observed.

Engineering exercise

An analyst fills a three-day reporting gap with zero admissions because the table contains blank cells. A trained model learns a fabricated collapse and rebound. The safer design distinguishes observed zero from unavailable data, carries a completeness marker and either imputes under a documented rule or rejects the issue-date forecast.

Apply the lesson

01	Define five core features with formulas and time-availability rules.
02	Create tests for duplicate geography-week keys and future-release records.
03	Run a validation comparison with and without one noisy covariate.

IN THE REPORT	Model inputs reflect the data actually available at the issue cut-off, including explicit missingness.

PART III / FORECASTING AS A TESTABLE CLAIM

CHAPTER 14 / INTERMEDIATE

Forecast targets, lead times and granularity

Design a prediction that can be scored fairly.

THE ANALYTICAL PROBLEM

A team announces that its model "predicts the next month" without clarifying whether it predicts a single monthly total, each future week, or the peak during the month. Those are not interchangeable targets. Operational users often need a sequence of weekly distributions, whereas a research paper might report only the fourth-week point error. Definition precedes architecture because it determines training labels, loss functions and reporting obligations.

Concept and definition

Use target = (variable, location, time unit, horizon, truth definition). A horizon-1 weekly forecast made on Friday may target the following complete week, the current incomplete week, or the next reporting release; spell out which. Counts and rates must not be silently mixed. Consider whether the spatial unit is a hospital, district, state or nation. Aggregation reduces some noise but can hide local overload. A multi-horizon model predicts several future steps in one invocation, so each step needs explicit calendar alignment and scoring.

CONCEPT MAP



Granularity
Temporal and geographic resolution of a target.

Multi-horizon
Producing forecasts for several future time steps.

Truth vintage
Version of realized target values selected for evaluation.

Evidence and method

Prepare a target specification with an example of a valid forecast row and a valid observed-truth row. Define the issue schedule, allowed reporting cut-off, truth vintage for scoring and any correction window. Choose horizons that match decisions: one-week tactical staffing, four-week procurement and longer-term exploratory scenarios may need different methods. If hospital definitions change during an evaluation season, report the break rather than comparing incompatible targets. Include who approves the target definition and the data owner responsible for revisions.

The expert perspective

At expert level, optimize a decision-relevant objective but continue reporting standard metrics. A model that minimizes average error nationally may underperform in high-risk smaller districts. Direct forecasting of multiple horizons can avoid recursive error accumulation, but it can also produce inconsistent trajectories if constraints are ignored. Aggregate forecasts should be reconciled with subregional forecasts when planners compare both levels. There is no single "best horizon" independent of the use case or measurement delay.

WATCH OUT	A forecast without a precise time target can be impossible to evaluate.

Analytic application

In our worked example, each Wednesday issue predicts admissions in each of the next four full reporting weeks, for one defined district. The prediction is a matrix with four rows and several quantiles per row, not merely a sentence saying "upward trend." A separate threshold summary can estimate capacity risk from the predictive distribution.

Apply the lesson

01	Write a machine-readable target row with dates, units and horizon.
02	Show how a different issue weekday changes the definition of "next week."
03	Specify the final truth source and how late revisions affect scoring.

IN THE REPORT	Four weekly horizon-specific distributions are issued using the designated reporting calendar and truth definition.

SELECTED SOURCE / [05] US Centers for Disease Control and Prevention

CHAPTER 15 / INTERMEDIATE

Simple baselines are formidable competitors

Learn persistence, seasonal naive and autoregressive thinking.

THE ANALYTICAL PROBLEM

A large Transformer can beat an uninformed guess while failing to outperform a two-line baseline. The second comparison is the one that matters. Seasonal epidemics often show strong persistence; near-term admissions may be reasonably approximated by recent values. A complicated model should earn its place through repeated out-of-sample improvement, not by producing polished-looking curves.

Concept and definition

A persistence forecast repeats the latest valid observation. A seasonal naive forecast repeats the corresponding prior seasonal period where comparable history exists. A moving average smooths noise, but it also responds slowly to turning points. Regression on recent lags, ARIMA-type methods and exponential smoothing add flexibility while retaining interpretability. None of these baselines can foresee an unprecedented shock, but they expose whether a supposedly powerful model is extracting real information rather than memorizing local trends.

CONCEPT MAP



Persistence
Forecast repeating the most recent observation.

Seasonal naive
Forecast repeating an analogous seasonal observation.

Forecast skill
Performance improvement relative to a named reference method.

Evidence and method

Build at least three baselines before training a neural system. Freeze their hyperparameters through the same rolling-origin protocol used for the Transformer. Evaluate them by horizon, jurisdiction and epidemic phase, not only with one overall mean. Inspect where they fail: during a rapid upswing, persistence may systematically underpredict; near a peak, linear trends may overshoot. A baseline should be simple enough to reproduce, clearly defined and available as a fallback when the primary model fails validation.

The expert perspective

Advanced comparisons should include a modest statistical model and a deliberately constrained neural model. Equalize available information across candidates: do not let the Transformer ingest a future calendar annotation that the baseline is denied, or give the baseline unrevised truth while the neural model sees issue-time data. Forecast skill is relative to a reference and can be negative. Investigate model selection instability across seasons and avoid selecting only the window favorable to the preferred architecture.

WATCH OUT	Beating a flat line on one chosen week proves very little.

Analytic application

If admissions in the latest eligible week are 202, the persistence baseline predicts 202 at each of four future weekly horizons. A richer model might produce medians 220, 235, 247 and 255, but those numbers are not evidence of superiority. Only forecasts issued at earlier unseen origins, scored against later observed outcomes, can establish improvement.

Apply the lesson

01	Implement a last-observation persistence baseline.
02	Create a seasonal comparison using a comparable previous period.
03	Report model skill relative to the baseline, including negative results.

IN THE REPORT	The Transformer is retained only where out-of-sample probabilistic and decision metrics justify its complexity.

SELECTED SOURCE / [07] Hyndman & Athanasopoulos · [05] US Centers for Disease Control and Prevention

CHAPTER 16 / INTERMEDIATE

Disease mechanisms and changing regimes

Know when a historical pattern can stop being relevant.

THE ANALYTICAL PROBLEM

The next epidemic wave is rarely a clean repetition of the last. Immunity changes, a pathogen evolves, care pathways adapt and surveillance definitions are revised. A time-series model that learned a stable weekly cycle can fail precisely when the situation becomes important. Domain knowledge is therefore a partner to data-driven forecasting, providing hypotheses about what might change and warning signs for extrapolation.

Concept and definition

Mechanistic epidemiological models express relationships such as susceptible depletion, exposure, infection and recovery. Statistical models learn associations between observed features and future outcomes. Both make assumptions, but in different places. A Transformer can represent nonlinear relations and long dependencies; it cannot guarantee sensible behavior outside its training distribution. Distinguish the epidemic process from the observation system: a break in testing access may change the case series without changing infections at the same rate.

CONCEPT MAP



Regime shift
Change in the mechanism generating observed data.

Mechanistic model
Model built around assumed processes rather than associations alone.

Transportability
Ability to use learned relations in a different context.

Evidence and method

Create a regime-change register recording new variants, interventions, testing policies, treatment changes, reporting system transitions and unexpected data outages. Attach time-stamped evidence and avoid using a later formal announcement to code the change before it was known. Review forecast residuals and errors around each break. Consider robust alternatives: ensemble with statistical baselines, regime indicators known by issue time, narrower publishing conditions or explicit scenario analysis. A model should escalate unfamiliar conditions rather than silently extending historical confidence intervals.

The expert perspective

Expert research can combine neural models with mechanistic priors, hierarchical pooling and Bayesian state estimates. It can also test transportability: does a model trained across regions learn disease dynamics or mostly reporting conventions? Dataset shift may affect covariates, the outcome mechanism or both. No clever architecture fully resolves unidentified confounding or radically novel outbreaks. Consider stress tests that replace missing feeds, delay releases and alter policy assumptions. Forecast intervals calibrated on quiet periods may be too narrow during transitions.

WATCH OUT	The most consequential errors may occur precisely when history becomes least representative.

Analytic application

Suppose a region changes admission coding at the start of a new season. The series jumps 18% overnight. A naive forecaster treats the jump as rising disease incidence; a guarded system tags the definition change, seeks bridging statistics and temporarily qualifies or suspends comparisons across the break.

Apply the lesson

01	Mark one historical policy or definition break on a series plot.
02	Contrast a mechanistic hypothesis with a pure statistical explanation.
03	Draft a contingency statement for model failure during novel conditions.

IN THE REPORT	A recent reporting change limits comparability; scenario ranges are emphasized until new data validate the trend.

SELECTED SOURCE / [03] World Health Organization

CHAPTER 17 / INTERMEDIATE

A forecast is a distribution

Understand uncertainty, quantiles, calibration and threshold risk.

THE ANALYTICAL PROBLEM

A briefing says "255 admissions in four weeks." A ward manager needs to know whether 255 is the expected value, median, upper bound or merely an illustrative scenario. They also need to know how often the model has been wrong by a meaningful amount. A single point can be useful for bookkeeping, but a decision depends on the range and consequences of possible outcomes.

Concept and definition

A predictive distribution assigns probability to possible future values. Quantiles describe cut points: the 0.10 quantile has about 10% of predicted outcomes below it, under the model. An 80% central prediction interval can be formed from the 0.10 and 0.90 quantiles if those quantiles are coherent. Width and coverage are different: a very wide interval can cover many observations without being useful; a narrow interval may be misleadingly confident. Calibration asks whether realized outcomes match advertised probabilities across many forecasts.

CONCEPT MAP



Calibration
Agreement between predicted probabilities and observed frequencies.

Pinball loss
Asymmetric error score used for quantile predictions.

WIS
Proper scoring rule for collections of predictive intervals and medians.

Evidence and method

Train quantile models with pinball loss or construct distributions using a suitable probabilistic method. Score with weighted interval score (WIS), interval coverage, median error and, where appropriate, calibration diagnostics. Inspect each horizon and location. Derive a threshold probability from an actual predictive distribution or forecast samples; do not infer a precise exceedance probability from only three quantile points. A decision report should distinguish model uncertainty, incomplete surveillance, and explicitly imposed scenarios. Include an accessible explanation of what an 80% interval does and does not mean.

The expert perspective

Advanced teams may use conformal methods under defensible exchangeability assumptions, hierarchical reconciliation and distributional mixtures. Temporal dependence and regime change can invalidate naive calibration guarantees. A zero-inflated or count-valued outcome may require distribution-aware modeling and nonnegative constraints. Decision thresholds should be accompanied by costs of false alarms and misses, not embedded into the loss function without consultation. Quantile crossing, where a predicted lower quantile exceeds an upper one, is a model defect that needs prevention or repair.

WATCH OUT	An 80% interval is a statistical forecast claim, not an 80% certainty about one specific hospital decision.

Analytic application

For the synthetic four-week example, a hypothetical model reports week-4 median 255 admissions with an 80% central interval of 173 to 371. The interval signals material uncertainty: preparations might be justified, but 255 is not a guaranteed outcome. A statement such as "53% chance of exceeding 250" would require a fuller distribution or forecast samples, not just these three quantiles.

Apply the lesson

01	Translate 0.10, 0.50 and 0.90 quantiles into ordinary language.
02	Check interval coverage on held-out forecast origins.
03	Explain why a threshold probability cannot be deduced exactly from three quantiles.

IN THE REPORT	The week-4 median is 255, with an 80% central interval of 173-371 admissions in the synthetic demonstration.

SELECTED SOURCE / [11] Bracher, Ray, Gneiting & Reich

CHAPTER 18 / INTERMEDIATE

Backtesting without time travel

Simulate the information constraints of historical issue dates.

THE ANALYTICAL PROBLEM

A random train-test split works for many independent examples. It is dangerous for forecasting, where neighboring weeks are related and later records contain information about earlier events. Randomly distributing observations lets the model learn from the future. The remedy is an evaluation scheme that recreates what an analyst could have known at each historical publication date.

Concept and definition

Rolling-origin backtesting moves the forecast issue date forward through a sequence of historical cut-offs. At each origin, train only on allowed past data, prepare features from available releases and forecast the designated future horizons. Once targets are realized, score the issued forecasts. A single held-out season is informative but can be unusually easy or hard. Across many origins, compare performance in growth, peak, decline and low-incidence periods. Geographic splits are also needed when claiming generalization to unseen regions.

CONCEPT MAP



Rolling origin
Chronological validation with advancing issue dates.

Look-ahead bias
Advantage from data unavailable at the historic forecast time.

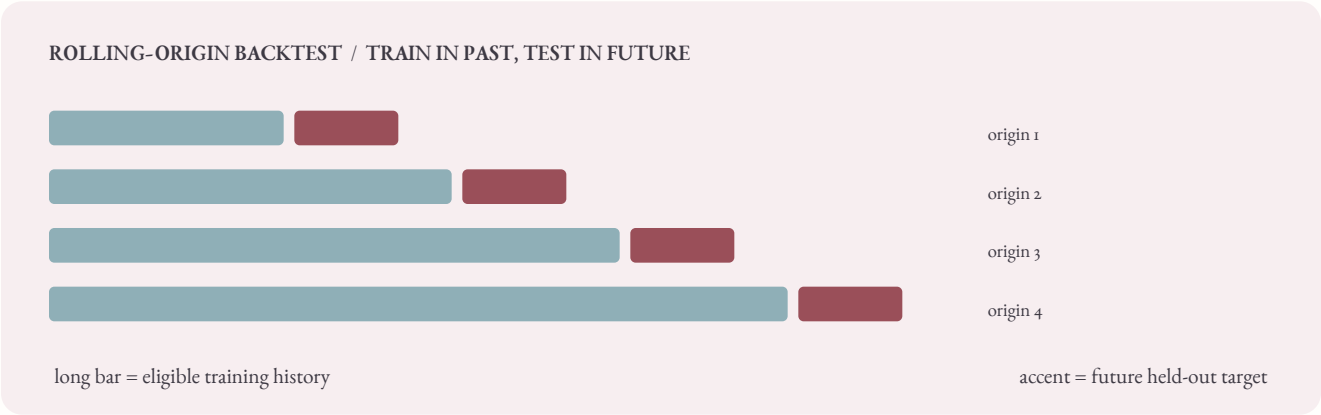
Shadow deployment
Running a model without using its outputs for live decisions.

Evidence and method

Specify a backtest protocol before model tuning: origin dates, training window, minimum history, retraining frequency, feature availability query, model selection interval and frozen final test span. Distinguish validation from test; the test should not be inspected repeatedly to tune architecture choices. Fit all scalers and imputers within each training fold. When data revisions matter, recreate vintage-specific inputs and use a declared final-truth scoring policy. Record missing forecasts separately from scored failures so a model cannot appear better by abstaining only on difficult weeks.

The expert perspective

Advanced evaluation includes paired score differences across matching origins, uncertainty about estimated skill, and hierarchical analysis by jurisdiction. Examine dataset leakage through target transformations, normalization, encoded event labels and retrospective source enrichment. A model re-trained every week must be compared against baselines with the same access schedule. For high-stakes use, a shadow period before live release reveals operational failures that an offline evaluation cannot measure. Report the distribution of scores, not only one attractive mean.



WATCH OUT A historical test is valid only when it obeys historical information availability.

Analytic application

Imagine backtesting a forecast issued on 1 February using a table downloaded on 1 June. If that table contains cases revised on 15 February, the earlier model now receives information from the future. Rebuilding the 1 February source vintage may reduce apparent accuracy, but it produces a truthful evaluation.

Apply the lesson

01 Draw three rolling-origin train and test windows.

02 List every step that must be fit separately inside a training fold.

03 Identify one way a retrospective text-event extraction could leak future facts.

IN THE REPORT Backtest results were computed over pre-specified rolling origins with issue-time source vintages.

SELECTED SOURCE / [07] Hyndman & Athanasopoulos

CHAPTER 19 / ADVANCED

What accuracy means to a decision-maker

Compare models using losses, phases and operational outcomes.

THE ANALYTICAL PROBLEM

A model with lower average absolute error can still be worse for a hospital director if it consistently underestimates fast-moving surges. Another model may be slightly less accurate on average but more honest about uncertainty. Model selection is a multi-criteria decision: statistical skill, calibration, robustness, explainability, operational cost and expected consequences all matter.

Concept and definition

MAE is the average absolute distance between point forecasts and realized outcomes. RMSE emphasizes larger errors. WIS rewards sharp, well-calibrated probabilistic forecasts and penalizes observations outside claimed intervals. Coverage checks the observed fraction inside an interval against its nominal level. Threshold-oriented recall asks how often a model identifies events above an operational level, but it must be considered alongside false alarms. These measures answer different questions and cannot simply be averaged without a stated objective.

CONCEPT MAP



MAE
Mean absolute error of point forecasts.

RMSE
Root mean squared point error, emphasizing large misses.

Decision loss
Cost of actions and misses under a stated policy.

Evidence and method

Construct an evaluation report with overall metrics and stratification: horizon, geography, epidemic phase, volume and data completeness. Compare against persistence, seasonal naive, a reasonable statistical forecaster and the chosen Transformer. Include missing-output rates and computation requirements. Use predeclared performance gates for deployment, such as no material reduction in coverage and improvement over baseline on WIS across relevant horizons. If thresholds are operationally important, perform decision analysis under several plausible false-positive and false-negative costs.

The expert perspective

Experts should quantify ranking uncertainty. Small validation samples often make model leaderboard positions unstable. A change in observation definitions can dominate small score differences. Use paired comparisons because every model should face the same forecast origins. Examine worst-case geographic gaps, not just a pooled national mean. Human reviewers may prefer an ensemble whose performance degrades gradually under input failures to one brittle architecture with the very best average score. Performance claims belong to specified datasets and periods; they are not universal endorsements of an architecture.

WATCH OUT	A single leaderboard number cannot capture the quality of an operational forecast.

Analytic application

In a purely illustrative held-out comparison, the persistence baseline has WIS 49, a simple statistical model 45 and a Transformer 43. The Transformer is better on this particular score, but a reviewer also sees its 80% interval coverage is only 69%. That coverage defect may be important enough to delay deployment or recalibrate the system, even though the point-ranking looks attractive.

Apply the lesson

01	Create a table of at least three metrics and two operational failure measures.
02	Explain a case where RMSE and WIS lead to different model choices.
03	Define a deployment gate using coverage and a baseline-relative score.

IN THE REPORT	The candidate improves median scoring but requires calibration review before it can guide high-stakes action.

SELECTED SOURCE / [11] Bracher, Ray, Gneiting & Reich · [05] US Centers for Disease Control and Prevention

PART IV / TRANSFORMER FORECASTING IN PRACTICE

CHAPTER 20 / FOUNDATION TO ADVANCED

From linear algebra to neural forecasting

Build enough mathematical intuition to read Transformer designs.

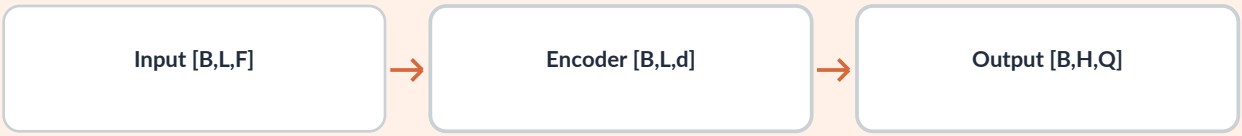
WHY THIS COMPONENT EXISTS

The word "Transformer" can make forecasting sound mysterious. At its foundation, the model moves arrays of numbers through learned transformations. A sequence of weekly observations becomes a matrix; neural layers transform that matrix into representations from which future values are predicted. Readers do not need to memorize every derivative to understand what the model can and cannot learn. They do need to recognize dimensions, loss functions and the boundary between training and inference.

Core contract

Suppose B is the batch size, L the look-back length, F the number of input features, and H the forecast horizon. The input tensor has shape [B, L, F]. A learned projection maps features to a hidden width d, creating [B, L, d]. Attention mixes information across positions or variables, depending on the architecture. An output head produces H future values or H times Q quantiles. Training adjusts weights to reduce a loss on historical labeled windows. Inference freezes those weights and uses only available past inputs.

CONCEPT MAP



Tensor
Multidimensional array of numerical values.

Embedding
Learned representation in a hidden feature space.

Loss function
Quantity minimized during model training.

Implementation contract

Start with a tiny model predicting four future admissions weeks from twelve past weeks. Standardize numerical features using training data only. Draw a window diagram: weeks t-11 through t form the input, weeks t+1 through t+4 form the label. Verify that the same week never appears as both an unavailable future input and a target. Print tensor shapes at every layer. Overfit a tiny synthetic batch to ensure the code can learn at all, then test whether it generalizes to an unseen chronological block.

The expert perspective

An expert will inspect inductive biases, regularization, model capacity and optimization. Forecasting data often provide far fewer independent examples than their row count suggests because overlapping windows share observations. Bigger networks can memorize seasons and regions rather than learn generalizable dynamics. Normalization can leak future statistics; nonstationarity changes feature distributions; high-dimensional covariates can drown the target history. Mathematical sophistication should therefore accompany experimental restraint. A linear model or modest multilayer perceptron is a useful reference when evaluating every architectural innovation.

WATCH OUT	A neural model that memorizes training windows is not a validated forecasting system.

Engineering exercise

For B=32 regions-times-origins, L=16 weeks, F=12 features and H=4 horizons, a batch input is [32,16,12]. If the model predicts 0.10, 0.50 and 0.90 quantiles, the final output is [32,4,3]. A mismatch such as [32,3,4] can silently corrupt loss computation if broadcasting happens unexpectedly.

Apply the lesson

01	Sketch the input and output tensor shapes for one target.
02	Identify the training split in which scaling parameters may be estimated.
03	Explain why 10,000 overlapping windows are not 10,000 independent epidemics.

IN THE REPORT	Inputs cover sixteen available weeks and output four future weekly distributions, without using future observations.

SELECTED SOURCE / [12] Vaswani et al.

CHAPTER 21 / INTERMEDIATE

Self-attention, step by step

See what queries, keys and values actually do.

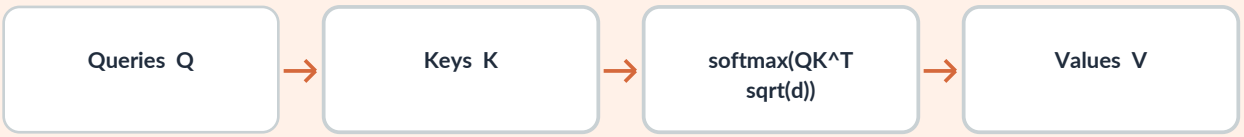
WHY THIS COMPONENT EXISTS

Imagine trying to predict admissions after several weeks of irregular activity. The last value matters, but so might a similar increase six weeks ago. Self-attention gives the model a way to combine information from different positions based on learned relevance. It is not a magic epidemiological explanation: it is a weighted computation over representations. Understanding that distinction makes Transformer papers much easier to read critically.

Core contract

Each input representation is projected into query, key and value vectors. Similarity between a query and available keys produces attention scores; after scaling and softmax, the scores act as weights over values. In standard notation, Attention(Q,K,V) = softmax(QK^T / sqrt(d_k)) V. Multiple heads can learn different representation patterns. Positional information is necessary when the architecture otherwise treats the same tokens similarly regardless of order. The exact interpretation depends on whether tokens correspond to time steps, patches or features.

CONCEPT MAP



Query-key-value
Learned projections used to compute attention.

Softmax
Normalization mapping scores into positive weights summing to one.

Mask
Restriction that prevents attention to invalid positions.

Implementation contract

Build intuition with three historic weeks rather than a 200-layer network. Write down one query and three similarity scores, normalize them to weights and compute the weighted value sum. Ask which tokens are accessible: an encoder using the full observed look-back can attend across that history, while an autoregressive decoder must be masked against future targets. A missingness or padding mask has a different role: it prevents fabricated padded steps from influencing representations. These details determine whether model training faithfully matches real forecasting.

The expert perspective

Experts should resist treating attention weights as causal explanations. They can reveal which internal representations were emphasized under one parameterization, but alternative weight patterns may yield similar outputs. Computational cost also matters: ordinary full attention across L temporal tokens has quadratic attention-matrix growth in L. Patch methods reduce token count; inverted methods attend over feature variables instead. Evaluate these trade-offs against data size and horizon instead of accepting a blanket claim that attention solves long-range forecasting.

WATCH OUT	Attention is a model mechanism, not a substitute for source validation or causal reasoning.

Engineering exercise

Suppose a query assigns normalized weights 0.2, 0.5 and 0.3 to three value vectors representing earlier periods. The resulting representation is their weighted sum. This describes one computation, not "a 50% chance that week 2 caused the forecast." The latter would be a causal and probabilistic claim unsupported by the attention weights.

Apply the lesson

01	Write the attention formula and explain each symbol in words.
02	Draw where a causal mask is required and where it is not.
03	Explain why the highest attention weight is not necessarily the strongest causal driver.

IN THE REPORT	Historical patterns receive learned weights within the forecasting model, but those weights are not treated as causal evidence.

SELECTED SOURCE / [12] Vaswani et al.

CHAPTER 22 / ADVANCED

Sequences, patches and future covariates

Choose what a Transformer token represents.

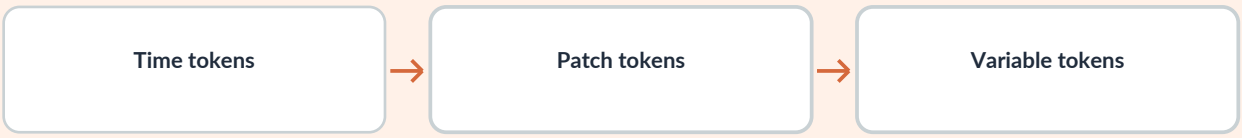
WHY THIS COMPONENT EXISTS

A time-series Transformer must answer a design question that language models mostly inherit from their task: what is a token? It could be one week containing twelve measurements, a small patch of consecutive weeks, or an entire history of one variable. Each choice privileges different relations and affects cost, interpretability and robustness to missing sources. The architecture should reflect the available data, not only a fashionable diagram.

Core contract

A temporal-token model embeds all feature values at each time step and attends across steps. A patch model groups adjacent steps before attention, retaining local motifs while shortening the token sequence. Inverted designs may encode each variable's historical trajectory as one token and attend across variables. Future covariates require discipline: holidays and known reporting calendar features may be truly available; future lab positivity and admissions are not. Some models accept separate historic observed, future known, and static feature groups.

CONCEPT MAP



Patch
Contiguous segment of time steps treated as a learned token.

Temporal token
Representation associated with one observation time.

Channel
Individual variable within a multivariate series.

Implementation contract

Draw the tensor contracts for each design. For time tokens, [B,L,F] becomes [B,L,d]. With nonoverlapping patches of length P, a single channel may become roughly L/P tokens, subject to padding and stride. For variable tokens, the token count often follows F rather than L. Test how missing values and changes in feature roster affect the chosen representation. Define exactly how the model emits H future outputs, and whether it predicts directly or autoregressively. Include ablations for position information, context length and covariate types.

The expert perspective

Experts compare memory, sample efficiency and inductive bias. Long attention windows are not automatically valuable if the signal is seasonal but source definitions change annually. Patch boundaries may obscure abrupt events; channel independence may underuse cross-series interactions; variable-token methods may struggle when features have very different semantics. Non-temporal documents should be fused through verified embeddings or events, not appended as arbitrary "weeks." Decoding choices also affect error propagation and interval coherence across horizons.

WATCH OUT	Passing actual future observations as "future covariates" invalidates evaluation.

Engineering exercise

Consider a 24-week, 10-feature history. A temporal-token encoder begins with 24 tokens. Six-week nonoverlapping patches yield four tokens per channel in one patch design. A variable-token encoder may instead start with ten tokens representing ten feature histories. These are alternative representations, not simple hyperparameter swaps.

Apply the lesson

01	Specify the token and tensor shape for two architecture families.
02	Label each proposed covariate as known-future, observed-past or static.
03	Predict which representation might be more robust to a missing feature.

IN THE REPORT	The tokenization design is documented and tested against simple alternatives over the same input history.

SELECTED SOURCE / [12] Vaswani et al.

CHAPTER 23 / ADVANCED

Temporal Fusion Transformer

Understand a design built for heterogeneous covariates and horizons.

WHY THIS COMPONENT EXISTS

Pandemic forecasting combines types of data that should not be treated identically: static regional context, past observed indicators, and calendar information known into the future. Temporal Fusion Transformer (TFT) was designed for this general multi-horizon setting. It is attractive not because its attention mechanism certifies scientific explanation, but because it explicitly distinguishes input roles and learns to select and transform variables.

Core contract

TFT, described by Lim and colleagues, combines static covariate encoders, variable-selection networks, gating components, sequence modeling and interpretable temporal attention. Static features contextualize a series; observed historical covariates inform past conditions; known-future features provide future calendar context. Variable selection changes with examples and learned representations. The model can produce multi-horizon quantiles, which align with report-ready uncertainty ranges. Its complexity requires sufficient training data and careful feature semantics.

CONCEPT MAP



TFT
Temporal Fusion Transformer for mixed-covariate multi-horizon forecasting.

Gating
Learned mechanism that suppresses or controls transformed information.

Static encoder
Component incorporating time-invariant contextual features.

Implementation contract

For a pandemic admissions project, organize tensors into static (region type, age structure), historical observed (admissions, test positivity, ED activity), and known future (week-of-year, fixed holidays). Build a documented data adapter that enforces this separation. Compare TFT with persistence, a seasonal model and a simpler neural baseline on identical rolling origins. Record both WIS and empirical interval coverage. Use TFT variable-selection and attention diagnostics as candidate hypotheses for analyst review rather than auto-generated causal narratives.

The expert perspective

An expert review inspects architecture assumptions, missingness treatment, feature selection stability and computational costs. Gate components can suppress irrelevant information, but that does not guarantee robustness under unseen data-generating regimes. Importance diagnostics may shift when correlated indicators substitute for one another. Test whether the model learns clinically implausible dependence on future-known metadata that proxy for reporting artifacts. Evaluate forecast quality separately for short and long horizons; a multi-horizon architecture can excel on one and fail on another.

WATCH OUT	Interpretability mechanisms are diagnostic aids, not causal validation.

Engineering exercise

A city-level TFT receives sixteen weeks of admissions and emergency-visit activity, static capacity and age-structure descriptors, and known week-of-year. It predicts four weekly quantile sets. If a policy announcement is unavailable at issue time, it stays out of known-future inputs regardless of how useful it appears in a retrospective study.

Apply the lesson

01	Sketch the three TFT feature groups and give two examples of each.
02	Describe what a variable-selection diagnostic can and cannot establish.
03	Compare TFT against a simpler model using horizon-specific WIS.

IN THE REPORT	TFT generates quantile forecasts using clearly separated historical, static and known-future inputs.

SELECTED SOURCE / [08] Lim, Arik, Loeff & Pfister

CHAPTER 24 / ADVANCED

PatchTST and long-context learning

Learn why patching can make attention efficient.

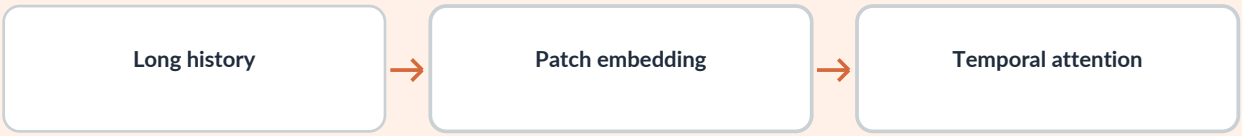
WHY THIS COMPONENT EXISTS

A full-attention model that ingests years of daily observations can become expensive, even when many neighboring days convey similar local information. PatchTST proposes a different unit of representation: short subsequences become tokens. The approach is conceptually accessible to a beginner who understands sliding windows and useful to experts studying long context lengths.

Core contract

In PatchTST, a time series is segmented into patches, with Transformer operations on patch embeddings. The paper emphasizes channel independence: different univariate series can share embedding and Transformer weights. Patching preserves local patterns inside tokens and reduces the number of attention positions compared with a token per observation. The design has demonstrated strong results on long-horizon forecasting benchmarks. This is not a guarantee of superiority on short, noisy epidemiological panels.

CONCEPT MAP



PatchTST
Patch-based Transformer architecture for time-series forecasting.

Context length
Amount of historical data supplied to a forecast.

Stride
Step size between successive sequence patches.

Implementation contract

Before testing patches, define why long look-back may matter: recurring seasonal timing, slowly changing baseline capacity or multiwave history. Choose patch length, stride and context length together, because overlapping patches affect token count and feature resolution. Verify that any scaling is fitted on the training period. Compare against the same feature set in a simpler encoder. For mixed static and categorical context, design a separate fusion layer rather than assuming a channel-independent encoder can interpret every variable type.

The expert perspective

PatchTST's channel independence can help learning from heterogeneous channels but may underrepresent direct interactions between sources. Additional fusion, shared context embeddings or ensemble components may be necessary. Short event spikes can be diluted by coarse patching, and cross-location patterns require deliberate pooling. Experts should compare skill and computational cost at several context lengths, including a much shorter history. A large context window that adds no generalization value merely increases sensitivity to obsolete surveillance regimes.

WATCH OUT	A longer context is only useful when its older observations remain relevant and comparable.

Engineering exercise

For a two-year daily series, representing every day as a separate attention token creates hundreds of positions. Grouping days into patches can shorten the sequence. In a weekly hospital-admissions panel with only 30 observations, however, patching may destroy rather than reveal information. Representation choice depends on observed granularity and sample size.

Apply the lesson

01	Compare the token counts for step-wise and patched inputs.
02	Propose a patch-length ablation study with a short-context baseline.
03	Identify one important dependency that channel independence might miss.

IN THE REPORT	Patch-based forecasting is evaluated as a candidate for long history, not adopted solely on benchmark reputation.

SELECTED SOURCE / [09] Nie et al.

CHAPTER 25 / ADVANCED

iTransformer and architecture selection

Compare models by their assumptions rather than their novelty.

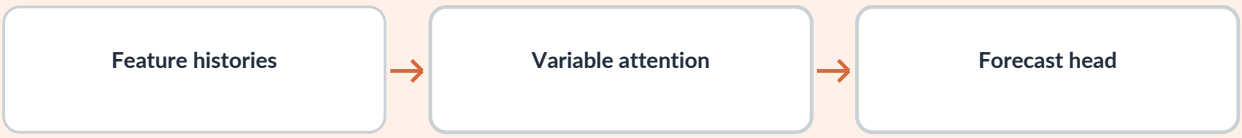
WHY THIS COMPONENT EXISTS

Two Transformers can use the same attention operation yet learn different relationships because the tokens stand for different things. iTransformer changes the common time-token perspective by embedding each variable's observed history as a token. This makes cross-variable relationships central to attention. Such a design could be interesting when a forecast uses admissions, lab positivity, syndromic visits and wastewater measurements, provided their availability and normalization are sound.

Core contract

In a conventional time-token design, one token may combine all variables at a particular week. In an inverted design, a token can represent the historic sequence of one variable. Attention then models relations across variates, while other transformations handle temporal representation within each variable. Published work reports competitive results across benchmark datasets. The operational question is narrower: does this design improve predictions on a health-surveillance dataset with missing feeds, changing measurement conventions and few independent epidemic seasons?

CONCEPT MAP



iTransformer
Architecture representing series variables as Transformer tokens.

Architecture ablation
Controlled test of a modeling design component.

Robustness
Ability to maintain acceptable behavior under perturbation or shift.

Implementation contract

Build an architecture decision record. Include dataset size, length of history, number and kinds of variables, forecast horizon, missingness, required latency and interpretability needs. Benchmark a classical model, a simple feedforward or linear neural model, TFT when feature roles are rich, a patch design for long history and an inverted design where cross-variable relations are a priority. Keep splits, available covariates and evaluation metrics identical. Measure training stability across random seeds rather than relying on one lucky run.

The expert perspective

At expert level, question whether attention across variables is scientifically meaningful when some channels are counts, others percentages and others media events. Sensible scaling and availability masks are essential. An apparent gain may result from using more recent data or a different training budget rather than architectural structure. Hold model selection to predeclared procedures, measure robustness to source withdrawal, and document failure cases. Sometimes the appropriate result is that a simpler ensemble remains better than any Transformer.

WATCH OUT	Benchmarks outside public health do not establish performance on a pandemic forecast task.

Engineering exercise

An experiment finds that the inverted model slightly improves WIS overall, but deteriorates when a laboratory feed goes missing. The team reports both findings and might select an ensemble or implement robust missingness augmentation. A credible research note does not hide the second result to produce a cleaner success story.

Apply the lesson

01	Write selection criteria before running an architectural bake-off.
02	Test identical evaluation origins and data for all models.
03	Include a source-outage stress test for multivariate attention models.

IN THE REPORT	Model selection was based on matched rolling-origin results and robustness tests, not architecture novelty.

SELECTED SOURCE / [10] Liu et al.

CHAPTER 26 / ADVANCED

Joining time series and unstructured intelligence

Fuse numeric observations, text events and regional context responsibly.

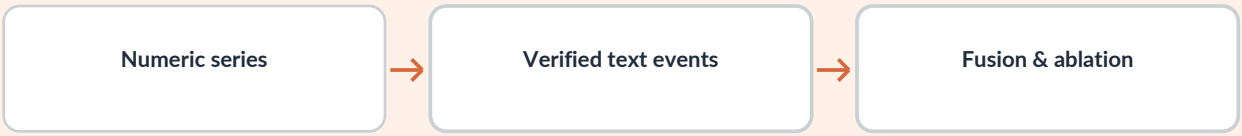
WHY THIS COMPONENT EXISTS

Suppose admissions rise just after a laboratory network changes its testing strategy. Weekly numeric features reveal a trend; a policy document explains why the trend may be difficult to interpret. A multimodal model could incorporate both. The challenge is to make the documents available at the correct time, distinguish statements from verified events, and keep the model from learning a spurious dependence on news volume.

Core contract

Multimodal fusion combines representations of distinct information types. The simplest approach encodes text-derived verified event counts or flags, joining them with quantitative weekly features. A more flexible approach embeds documents and pools them by jurisdiction and publication period. Static features can enter through learned embeddings or conditioning layers. The forecast target remains a measured numeric outcome. A separate retrieval system can provide documentary evidence to the report writer without requiring the forecasting model to consume raw prose.

CONCEPT MAP



Multimodal fusion
Combining distinct data modalities under explicit semantics.

Cross-attention
Attention between representations of different token sets.

Evidence retrieval
Finding passages that substantiate a report claim.

Implementation contract

Use a ladder of complexity. Begin with verified binary event features and compare performance with no event data. Next test aggregate event counts with source-volume controls. Only then consider learned document embeddings, ensuring that the exact text was published before each forecast origin. Deduplicate syndication, preserve multilingual evaluation examples, and audit whether sources overrepresent major cities. Use held-out origins to measure incremental forecast value. In reporting, retrieve supporting documents by claim identifier rather than trusting the generative model to remember what it read.

The expert perspective

Advanced architectures may use cross-attention between temporal states and document embeddings, though temporal and semantic alignment become difficult. Quantitative evaluation should include ablation of all text features, not only the preferred extraction model. Where document extraction changes over time, fix or version the NLP model during retrospective comparisons. Resist treating semantic similarity as proof that two events are equivalent. Build an explicit interface: probabilistic forecaster returns numerical distributions, evidence retrieval returns cited passages, and the report composer must not fabricate a bridge between them.

WATCH OUT	A semantic embedding does not encode truthfulness by itself.

Engineering exercise

An event extractor identifies a newly announced laboratory reporting rule. It records publication date, effective date and source span. A forecaster may use the event as an issue-time covariate; the report generator may cite the rule as a reason for interpreting current counts cautiously. Neither component should claim that the rule caused a quantitative change without additional analysis.

Apply the lesson

01	Compare no-text, event-flag and document-embedding variants in one backtest.
02	Check publication-time access for each selected document.
03	Keep numerical forecasts and text-supported explanations in separate output fields.

IN THE REPORT	Text-derived events supplement the quantitative forecast only when independently verified and available on time.

SELECTED SOURCE / [08] Lim, Arik, Loeff & Pfister · [09] Nie et al. · [10] Liu et al.

CHAPTER 27 / ADVANCED

Training, tuning and constrained outputs

Build a repeatable Transformer experiment instead of a demo.

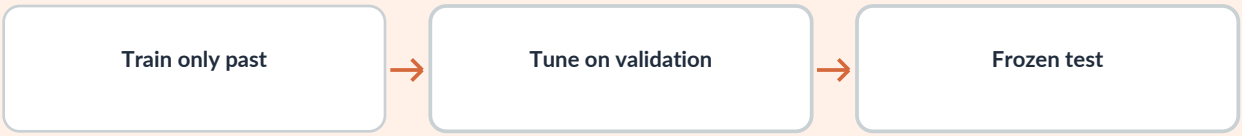
WHY THIS COMPONENT EXISTS

A training notebook that produces attractive lines is not yet a forecasting method. Small choices - where normalization was fitted, how weeks were split, which metric selected checkpoints and whether a missing value meant zero - can dominate the results. Reliable training is therefore more about controlled experiments than endlessly increasing parameter counts.

Core contract

A typical training loop draws historical look-back windows, computes horizon-specific predictions, compares them to realized future targets and updates weights through gradient descent. For quantile outputs, pinball loss is a natural objective. The model should respect nonnegative target support and ordered quantiles. Techniques such as dropout, early stopping and weight decay may reduce overfitting, but their value must be shown through validation. Keep random seeds, package versions, data vintages and configurations with every run.

CONCEPT MAP



Early stopping
Ending training when held-out validation performance stops improving.

Quantile crossing
Predicted higher quantile falling below a lower quantile.

Experiment registry
Versioned record of configurations, inputs, artifacts and scores.

Implementation contract

Begin with a small architecture and a data pipeline test suite. Verify shapes, masks and gradient flow on synthetic windows. Fit scalers using training data only, then construct chronological train/validation/test splits. Log learning curves and checkpoint by validation WIS or quantile loss, not test results. Stress-test missing data, short-history regions and late revisions. If the deployment schedule retrains weekly, reproduce that schedule in validation. For larger experiments, cap candidate searches and record every unsuccessful configuration rather than only the winner.

The expert perspective

At expert level, calibration and distribution coherence become central. Nonnegative monotone quantiles can be obtained through constrained parameterizations, while probabilistic likelihood models should match the count behavior. Temporal cross-quantile and cross-horizon inconsistencies may require further modeling. Compare a calibrated ensemble with each component, and inspect whether postprocessing degrades sharpness. Deep models are particularly vulnerable to selection on a small validation season. Adopt an experiment registry and formal approval gates so a claimed improvement can be reproduced by a second analyst.

WATCH OUT	Do not use the final test set as an informal hyperparameter tuning dashboard.

Engineering exercise

A model achieves low training error but predicts extremely narrow intervals on an unseen season. The correct response is not to publish the confident forecast. The team checks for leakage, compares validation and test coverage, evaluates calibration methods and, if necessary, falls back to a better-calibrated baseline.

Apply the lesson

01	Write tests for output shape, monotonic quantiles and nonnegative counts.
02	Keep a reproducible record of one training run and its data vintage.
03	Decide checkpoint selection before opening the final test results.

IN THE REPORT	The model passed shape, leakage, calibration and robustness checks under a frozen evaluation protocol.

SELECTED SOURCE / [08] Lim, Arik, Loeff & Pfister

PART V / FROM MODEL OUTPUT TO INTELLIGENCE REPORT

CHAPTER 28 / ADVANCED

Ensembles, explanation and dissent

Balance strong models with credible alternatives.

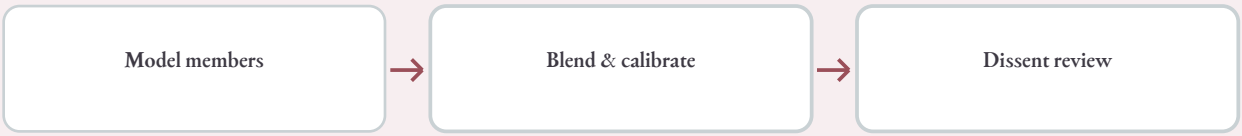
THE ANALYTICAL PROBLEM

A forecasting team naturally wants to select a winner. In practice, blending several competent but imperfect methods often produces more stable behavior. The role of an ensemble is not to make uncertainty disappear; it is to reduce dependence on a single modeling assumption. Analysts should also preserve dissenting signals, including the possibility that the trained model has entered an unfamiliar regime.

Concept and definition

An ensemble combines forecasts from multiple methods. Quantile averaging, mixture distributions and weighted aggregation are not equivalent; use methods compatible with the target representation and probabilistic interpretation. Compare the ensemble against its components on the same rolling origins. Explain forecasts by separating observable input trends, model outputs and analyst interpretation. Feature importance and attention diagnostics are model-dependent, while source lineage and timing are auditable facts.

CONCEPT MAP



Ensemble
Combination of forecasts from multiple models.

Model disagreement
Divergence among candidate predictions.

Common-mode failure
Multiple systems failing because they share an assumption or dependency.

Evidence and method

Design a forecast comparison sheet listing each member's latest median, interval, historical skill, current input health and known failure modes. Use weights selected on past validation evidence, not hand-adjusted after seeing the future target. If models disagree sharply, report the disagreement and investigate missing data or regime change. Include a dissent field in the publication checklist: "What would make our leading assessment wrong?" The purpose is disciplined skepticism, not perpetual indecision.

The expert perspective

Experts should consider dependence among ensemble members. Five Transformers trained on the same feature table may share failure modes, so their agreement offers less reassurance than it appears. Blend models with different inductive biases where evidence supports it. Examine calibration after aggregation rather than assuming every component's nominal coverage survives. Distinguish uncertainty from aleatoric variability, epistemic limitations and unsupported scenarios. Uncertainty labels should be traceable to an evaluation procedure.

WATCH OUT	Agreement among highly correlated models is not independent confirmation.

Analytic application

A seasonal baseline predicts 210 admissions in week four, TFT predicts 255 and PatchTST predicts 246. Rather than selecting the largest number as a precaution, the team checks historical skill, current source health and consistency. The report can show the ensemble range and note disagreement under late-reporting conditions. All numbers in this example are invented for teaching.

Apply the lesson

01	Compare a simple average to a weighted blend on historic origins.
02	Describe one common-mode failure across nominally different models.
03	Write a dissent sentence for an apparently confident forecast.

IN THE REPORT	Model outputs are broadly aligned; the remaining disagreement is included in the uncertainty assessment.

SELECTED SOURCE / [11] Bracher, Ray, Gneiting & Reich

CHAPTER 29 / ADVANCED

A production OSINT-to-forecast architecture

Turn the method into a secure, auditable operating system.

THE ANALYTICAL PROBLEM

A useful forecasting platform has to do the same thing again next week, when an API is late, one district changes its code and a model server restarts during briefing preparation. This is a software and operations problem as much as a statistical one. The solution is a sequence of explicit contracts rather than one enormous agent asked to collect, infer and publish everything in one pass.

Concept and definition

A maintainable design has acquisition adapters, an immutable raw store, validated standardized tables, a historical-vintage feature service, model training and inference services, an evaluation registry, evidence retrieval, report composition and approval. Each stage emits a versioned artifact. The generative language component should consume already validated forecasts and a citable evidence packet; it should not invent numerical predictions from raw articles. Strong boundaries make failures local and easier to diagnose.

CONCEPT MAP



Data contract
Explicit schema and semantics enforced between components.

Artifact lineage
Traceable relationship among inputs, models and outputs.

Abstention
Deliberate decision not to publish an unreliable forecast.

Evidence and method

Schedule source collection with retries, source-specific rate limits and integrity checks. Materialize only valid rows into standardized tables, recording provenance and warnings. Freeze the forecast issue-time snapshot, run baselines and candidate models, perform output sanity checks and generate a model card for the run. Assemble a report draft from structured JSON and selected verified evidence. A reviewer approves or returns it with corrections. Archive the approved PDF, machine-readable forecasts, model version and the exact source manifest. Use role-based access and security review appropriate to any sensitive holdings.

The expert perspective

Advanced teams separate live forecast generation from retraining so an unstable training job does not interrupt publication. An event bus or queue may help as volume grows, but simple scheduled jobs often suffice initially. Define service-level objectives for freshness, issue-time latency, missing output and rollback time. Model artifacts need cryptographic checksums and reproducible training environments. Automated narrative tests should reject statements whose figures disagree with structured model output or whose citations cannot be resolved.

WATCH OUT	The language model writes a report from checked outputs; it must not manufacture the forecast.

Analytic application

A Wednesday source feed fails. The pipeline refuses to mark last week's copied value as "fresh." It may issue a qualified baseline report, delay release or abstain, according to predefined policy. The report displays the missing feed and downgraded evidence status. That behavior is more professional than quietly hiding the incident in a smooth chart.

Apply the lesson

01	Sketch the raw-to-report pipeline with versioned artifacts.
02	Define a quarantine action when one input feed fails a schema test.
03	Specify what is archived with each published forecast.

IN THE REPORT	The release is reproducible from its source manifest, feature snapshot, model versions and approved report packet.

SELECTED SOURCE / [14] NIST

CHAPTER 30 / ADVANCED

Drift, incidents and human oversight

Keep the system reliable after the first successful release.

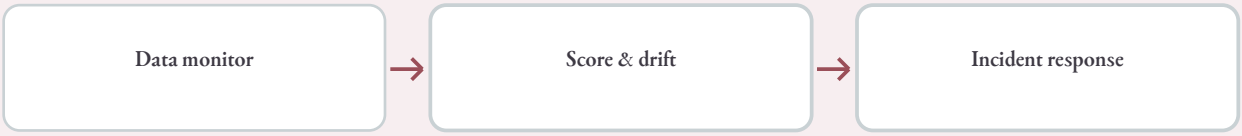
THE ANALYTICAL PROBLEM

A model passes evaluation, is deployed and then slowly loses its edge. Inputs become stale, seasonality shifts and a health authority changes definitions. Without monitoring, a polished dashboard can continue to present a degraded forecast as routine information. Production stewardship means detecting such changes and changing behavior visibly.

Concept and definition

Monitor data freshness, completeness, schema conformance, out-of-range values, model output validity, interval coverage and score against baselines. Some metrics are available immediately; error and calibration can be assessed only after outcomes arrive. Keep two distinct alert channels: an operational problem such as missing data, and a substantive epidemiological signal such as an unusual surge. Their response workflows differ. Human oversight should be a defined role with authority to qualify, suspend or correct a report.

CONCEPT MAP



Drift
Change in data or model performance over time.

Release gate
Automated or human condition required before publication.

Postmortem
Documented analysis of a failure and its corrective action.

Evidence and method

Create a runbook for failed collection, leaked feature discovery, model drift, unexplained forecast jumps and published-report errors. Each incident needs severity, owner, impact estimate, containment, correction, and postmortem. Monitor discrepancies between model outputs and narrative text before distribution. If the data definition changes, add a visible note and revalidate baselines. Run periodic reviews of source licenses, privacy risks and regional error inequity. A system that cannot revoke a bad report is not production-ready.

The expert perspective

An expert oversight committee reviews whether threshold alerts lead to useful actions and whether actions create feedback into future data. Measure cost of abstention as well as the cost of incorrect forecasts. Drift detectors are supplementary diagnostics; no single numerical threshold can declare all novel epidemic regimes. Maintain an independent analyst channel for unusual qualitative evidence that the quantitative model has not absorbed. Documentation and audit trails should make it possible to explain what changed between two report editions.

WATCH OUT A valid-looking model output is not proof that the upstream evidence is sound.

Analytic application

After a new lab-reporting policy, missingness in a leading indicator rises to 40%. The model still returns numeric quantiles. A monitoring gate recognizes the source outage, compares the run with a no-lab-input fallback and flags the briefing for review rather than trusting the successful API response as proof of forecast reliability.

Apply the lesson

01 List five operational checks and three retrospective scoring checks.

02 Write an incident response for a feature discovered to contain future information.

03 Define who can pause, approve or correct the published report.

IN THE REPORT The report is qualified because one leading data feed failed freshness and completeness checks.

SELECTED SOURCE / [14] NIST · [06] World Health Organization

CHAPTER 31 / FOUNDATION TO ADVANCED

Writing the prediction intelligence report

Translate distributions into defensible choices and clear prose.

THE ANALYTICAL PROBLEM

The final audience may never inspect a Tensor. They will read a one-page summary, a table of future outcomes and a sentence about what might change the assessment. Strong reporting does not hide rigor behind jargon. It provides a short, testable claim, names the evidence, expresses uncertainty honestly and links the prediction to an available decision. A report that omits the forecast issue date, units or sources is not complete, regardless of writing quality.

Concept and definition

A standard report includes title and issue timestamp, a decision question, key assessment, horizon-specific forecast table, interval chart, source and data notes, model method, past validation, scenarios, action thresholds and limitations. Use distinct headings for "Observed", "Predicted" and "Scenario" statements. For numerical predictions provide median and interval, not only adjectives. Discuss competing explanations for unusual signals. Keep methodological detail in an appendix while preserving the most important caveats near the summary where decision-makers will see them.

CONCEPT MAP



Key assessment
Short statement of the most decision-relevant conclusion.

Decision threshold
User-defined value at which an action may be considered.

Evidence packet
Selected, traceable source material supporting a report.

Evidence and method

Generate the draft from a structured forecast object: location, target, issue time, units, horizon rows, quantiles, baseline comparator, availability flags and model-version identifier. A separate evidence object contains verified claim IDs and source links. A text-generation model may help organize the prose, but use deterministic checks to verify that every figure in the text matches the structured table. Require citations for source-dependent claims. A qualified analyst reviews wording around probability, mechanism and recommended actions. Archive the final edition and changes.

The expert perspective

Expert report design calibrates communication for different readers without changing substance. Executives need material risks and decision options; model reviewers need scoring, leakage controls and definitions; public communication needs accessible uncertainty wording and privacy-safe geography. A threshold-crossing probability should come from a full predictive distribution, not invented by a language model. Some situations call for no forecast or for a bounded scenario statement. Credibility increases when the team reports what the model does not know.

WATCH OUT	Do not use a generative model to invent probabilities, sources or model scores.

Analytic application

A useful synthetic headline is: "Admissions may continue rising over the next four weeks, but the interval remains wide." The accompanying table gives each week's median and 80% range, the capacity review threshold, historical score and the caveat that recent surveillance is provisional. An action paragraph says what planners may wish to examine rather than pretending the model can mandate staffing policy.

Apply the lesson

01	Create a one-paragraph key assessment with target, horizon, uncertainty and caveat.
02	Check every narrative number against a structured forecast table.
03	Add a "what could change our view" paragraph with at least two triggers.

IN THE REPORT	The four-week outlook suggests rising demand, with wide uncertainty and a declared need for ongoing review.

SELECTED SOURCE / [13] Collins et al.

CHAPTER 32 / CAPSTONE

A complete synthetic district case

Follow the full chain from evidence to forecast and report.

THE ANALYTICAL PROBLEM

We finish with a fictional district called North Bay District. Every number in this case is invented for teaching, and none describes a real public-health situation. The decision is whether planners should review surge capacity over the next four weekly reporting periods. The selected observed target is weekly new admissions for a respiratory syndrome. The point of the case is not to prove that a Transformer succeeds; it is to show which artifacts make any forecasting claim inspectable.

Concept and definition

The last eight synthetic weekly admissions are 90, 104, 118, 136, 151, 169, 183 and 202. The issue-time source packet also contains a verified laboratory-reporting note, a provisional emergency-visit trend and static population context. The source note is not automatically a causal feature. The forecaster receives valid prior-week values only. A persistence baseline predicts 202 at all four horizons. A candidate Transformer issues illustrative median forecasts 220, 235, 247 and 255, with separate 0.10 and 0.90 quantiles.

CONCEPT MAP



Synthetic case
Educational illustration using invented rather than observed outcomes.

Forecast packet
Versioned set of validated predictions and metadata.

Decision-ready report
Output connecting evidence, uncertainty and possible action.

Evidence and method

The synthetic production protocol first freezes the evidence packet. It runs source-quality checks and builds the time-aligned feature table. It then executes the baseline and candidate model, validates ordered nonnegative quantiles and attaches the candidate's historical backtest record. Next it assembles a four-row table, interval visualization and a plain-language paragraph. A human reviewer checks the target definition and qualitative claims. The final report includes a publication timestamp, forecasts, data caveats, source manifest reference and triggers that would lead to reassessment.

The expert perspective

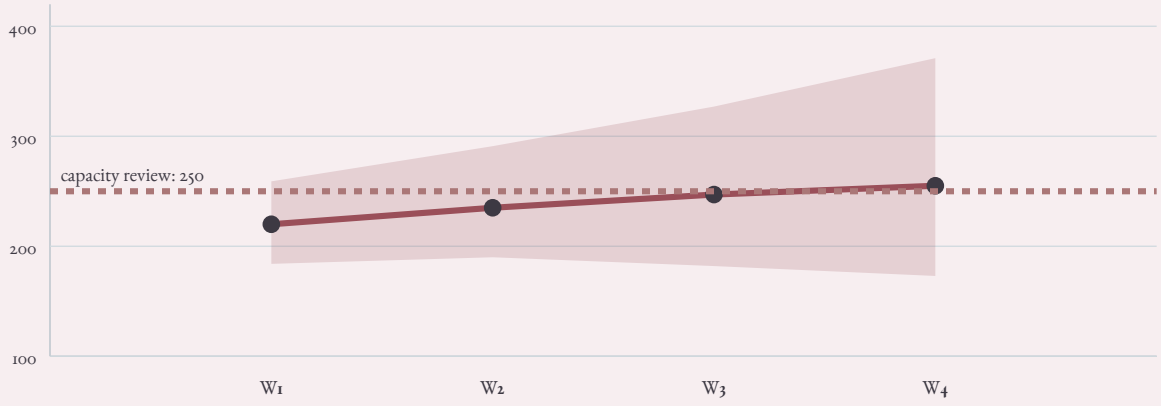
The advanced lesson is what remains unknown. The illustrative forecast does not estimate the effect of a hypothetical policy, establish the cause of increased admissions or yield an exact exceedance probability without a complete predictive distribution. If later published hospital data revise historical admissions, the team preserves the original issue-time packet while updating the evaluation truth according to policy. If the real system lacks sufficient past waves to validate the Transformer, the baseline may be the only defensible operational forecast.

WATCH OUT	All capstone data, metrics and predictions are fictitious and must never be cited as real pandemic evidence.

Analytic application

The fourth-week synthetic forecast is q10=173, median=255 and q90=371 admissions. The report can discuss the 250-admission capacity review level, but must not claim "a 53% probability of exceeding it" unless forecast samples or a distribution support that figure. The right conclusion is that materially different outcomes remain plausible and the user should use the range as one input to a capacity review.

FOUR-WEEK SYNTHETIC ADMISSIONS FORECAST



Shaded area: 80% central interval; all values fictitious

Apply the lesson

01 Reproduce the four-horizon table from the capstone data.

02 Write an executive paragraph that distinguishes observation, forecast and scenario.

03 Audit the capstone for time leakage, metric provenance and numerical consistency.

IN THE REPORT	Synthetic example: the week-4 median is 255 admissions, and the 80% interval spans 173 to 371.

APPENDIX A

Data contracts: a source becomes a forecast input

The following records are examples, not proposed universal standards. A real project should use the definitions, access rules and jurisdiction codes of its source owners.

A.1 Minimal source manifest

Field	Meaning	Example
source_id	Stable feed identifier	health-admissions-v2
source_uri	Original publication location	official-source.example/feed
retrieved_at	UTC ingestion instant	2026-10-08T09:10:00Z
published_at	Publisher release instant	2026-10-07T20:00:00Z
observation_end	Week ending date	2026-10-04
sha256	Hash of raw download	64 hexadecimal characters
definition_version	Target definition	admission-rule-2026.2
license_review	Documented permission	approved

A source identifier is not enough by itself. The publishing time, observation time and retrieval time answer three different questions. Archiving the file and recording a checksum makes later disagreements about revisions resolvable.

APPENDIX A / CONTINUED

A.2 Forecast object in JSON

A structured forecast should be the sole numerical authority for the report writer. Below is a compact educational packet for the fictional district.

```
{
  "forecast_id": "demo-north-bay-2026w41",
  "issued_at": "2026-10-08T12:00:00Z",
  "synthetic": true,
  "region": "North Bay District (fictional)",
  "target": "weekly new respiratory admissions",
  "unit": "admissions per week",
  "horizons": [
    {"week_ahead": 1, "q10": 184, "q50": 220, "q90": 259},
    {"week_ahead": 2, "q10": 190, "q50": 235, "q90": 291},
    {"week_ahead": 3, "q10": 182, "q50": 247, "q90": 327},
    {"week_ahead": 4, "q10": 173, "q50": 255, "q90": 371}
  ],
  "reference_model": "last-observation=202",
  "data_manifest_id": "demo-manifest-01",
  "model_id": "illustrative-transformer",
  "validation_status": "not empirically established"
}
```

IMPORTANT	This packet intentionally lacks a threshold exceedance probability. Three quantiles alone do not determine that probability.

APPENDIX A / CONTINUED

A.3 A clean feature contract

A feature is valid only when its values can be reconstructed from sources released by the intended issue date. The schema records both meaning and availability.

Feature	Data role	Availability rule
admissions_lag1	Observed past target	Released before issue cutoff
ed_visit_pct	Observed leading indicator	Latest eligible release only
calendar_week	Known future input	Deterministic for planned date
age_65_share	Static regional descriptor	Survey effective at issue date
policy_event_flag	Verified event	Published before issue cutoff
data_missing_flag	Quality metadata	Distinguish missing from zero
source_age_days	Freshness metadata	Issue minus latest release

Expert check: a policy announcement dated one week after a forecast origin does not belong in that origin's model input, even if its stated effective date falls earlier. The label "past" refers to information availability, not simply to the event being described.

APPENDIX B

A minimal Transformer you can inspect

The code below is an educational, self-contained PyTorch example. It uses invented synthetic values, predicts four future weeks and returns ordered nonnegative 10th, 50th and 90th percentiles. It is a simple Transformer encoder, not an implementation of TFT, PatchTST or iTransformer. It cannot be deployed for public-health decisions without real data, expert review and the validation described throughout this book.

SAFETY OF INTERPRETATION	Successful execution proves that the training code runs. It does not validate epidemiological accuracy, source quality, calibration or a capacity decision.

What to inspect

01	The input tensor contains sixteen available historic weeks and three features per week.
02	The output has four horizon rows and three ordered quantile columns.
03	Training, validation and test periods are separated chronologically; no future outcome enters past input windows.

APPENDIX B / SOURCE LISTING

B.1 Synthetic data and historic windows

```
"""Educational multihorizon Transformer on synthetic weekly data.
Not an epidemiological model or production-ready forecast.
Requires: numpy and torch. Run: python demo_forecast_transformer.py
"""
import math
import numpy as np
import torch
from torch import nn
from torch.nn import functional as F

torch.manual_seed(14)
rng = np.random.default_rng(14)
LOOKBACK, HORIZON, N = 16, 4, 240
WEEK = np.arange(N)
season = np.sin(2 * np.pi * WEEK / 52)
counts = np.maximum(0, 125 + 21*season + 0.23*WEEK
                    + rng.normal(0, 8, N))
# Three historic features: admissions, sine season, cosine season.
features = np.column_stack((counts,
                            np.sin(2*np.pi*WEEK/52),
                            np.cos(2*np.pi*WEEK/52)))
# Choose a fixed scale in this toy experiment; in real work fit a
# scaler using training origins only, then save its parameters.
features[:, 0] /= 200.0
scaled_counts = counts / 200.0

origin, X, Y = [], [], []
for cut in range(LOOKBACK, N-HORIZON+1):
    origin.append(cut)
    X.append(features[cut-LOOKBACK:cut])
    Y.append(scaled_counts[cut:cut+HORIZON])
origin = np.asarray(origin)
X = torch.tensor(np.asarray(X), dtype=torch.float32)
Y = torch.tensor(np.asarray(Y), dtype=torch.float32)
# Labels for train end by week 168; validation covers weeks 168-208.
tr = (origin + HORIZON <= 168)
va = (origin >= 168) & (origin + HORIZON <= 208)
te = (origin >= 208) & (origin + HORIZON <= N)
```

APPENDIX B / SOURCE LISTING

B.2 Attention encoder and ordered quantiles

```
class QuantileTransformer(nn.Module):
    def __init__(self, features=3, width=48, horizon=4):
        super().__init__()
        self.proj = nn.Linear(features, width)
        self.pos = nn.Embedding(64, width)
        layer = nn.TransformerEncoderLayer(
            d_model=width, nhead=4, dim_feedforward=96,
            dropout=0.10, batch_first=True, norm_first=True)
        self.enc = nn.TransformerEncoder(layer, num_layers=2)
        self.head = nn.Linear(width, horizon * 3)
        self.horizon = horizon

    def forward(self, x):
        B, L, F_in = x.shape
        idx = torch.arange(L, device=x.device)
        hidden = self.proj(x) + self.pos(idx)[None, :, :]
        representation = self.enc(hidden)[:, -1, :]
        raw = self.head(representation).reshape(B,self.horizon,3)
        # Monotone nonnegative q10 <= q50 <= q90.
        q10 = F.softplus(raw[..., 0])
        q50 = q10 + F.softplus(raw[..., 1])
        q90 = q50 + F.softplus(raw[..., 2])
        return torch.stack([q10,q50,q90],dim=-1)
```

APPENDIX B / SOURCE LISTING

B.3 Pinball loss, training and test coverage

```
def pinball(pred, truth):
    levels = pred.new_tensor([0.10,0.50,0.90])
    diff = truth[...,None]-pred
    return torch.maximum(levels*diff,(levels-1)*diff).mean()

model = QuantileTransformer()
optimizer = torch.optim.AdamW(model.parameters(), lr=0.002)
best_loss, best_state = float('inf'), None
for epoch in range(140):
    model.train()
    optimizer.zero_grad()
    pred = model(X[tr])
    loss = pinball(pred, Y[tr])
    loss.backward()
    optimizer.step()
    model.eval()
    with torch.no_grad():
        validation = pinball(model(X[va]),Y[va]).item()
    if validation < best_loss:
        best_loss = validation
        best_state = {k:v.clone().detach()
                      for k,v in model.state_dict().items()}
model.load_state_dict(best_state)
model.eval()
with torch.no_grad():
    q = model(X[te]) * 200.0  # back to admissions units
    actual = Y[te] * 200.0
    cover80 = ((actual >= q[:,:,0]) &
               (actual <= q[:,:,2])).float().mean().item()
print('Educational test interval coverage:',round(cover80,3))
print('First held-out origin q10/q50/q90 by horizon:')
print(q[0].round().int().tolist())
# Actual deployment additionally needs true source vintages,
# baselines, matched rolling-origin evaluation, scored intervals,
# current covariate availability and a human publication gate.
```

APPENDIX B / TESTING NOTES

How to turn the prototype into an experiment

The program creates seasonal toy data, partitions windows chronologically and estimates predictive quantiles. An operational adaptation needs an as-of data lake, exact epidemiological-week mapping, a count-compatible output treatment, source-wise missingness masks, and a published baseline comparator. Track real coverage and WIS across rolling origins rather than assuming the first held-out model is calibrated.

Three tests to write first

01	Temporal test: every source release timestamp in an input row is less than or equal to that forecast origin.
02	Tensor test: outputs have shape [batch, 4, 3], q10 <= q50 <= q90, and all predicted admissions are nonnegative.
03	Report test: quantile figures in every generated sentence exactly match the signed machine-readable forecast packet.

What the code deliberately omits

It does not perform vintage-aware historical reconstruction, clinical data verification, multilingual extraction, full WIS scoring, epidemic regime testing, hierarchical reconciliation, or release approval. These omissions are not hidden behind a "production-ready" claim. They define the next engineering tasks.

APPENDIX C / SAMPLE REPORT

A decision-ready pandemic intelligence brief

SYNTHETIC ILLUSTRATION / NORTH BAY DISTRICT / 8 OCTOBER 2026

STATUS	FICTIONAL EXERCISE. The location, historical values, model predictions and capacity review threshold are invented.

Decision question

Should district planners review surge capacity for the next four complete reporting weeks? The named operational indicator is weekly new respiratory-syndrome hospital admissions, reported by participating facilities in the fictional district.

Key assessment

The illustrative forecast points toward higher weekly admissions over the next four weeks, with increasing uncertainty at longer horizons. The model's week-4 median is 255 admissions and its central 80% interval is 173-371. The planning threshold of 250 admissions per week falls inside that interval. This is a risk-management prompt, not a verified prediction of real activity.

Observed context

The eight synthetic eligible weekly counts are 90, 104, 118, 136, 151, 169, 183 and 202. A persistence baseline therefore projects 202 at each future week. No claim about current real-world infection rates, cause of trend or interventions is implied. The toy series is deliberately simple so readers can inspect the reporting logic.

What could change the assessment

A source definition change, delayed facility reporting, abrupt changes in clinical severity, or previously unseen transmission patterns could alter the trajectory. Fresh source vintages and short-horizon realized admissions should be reviewed at each issue cycle.

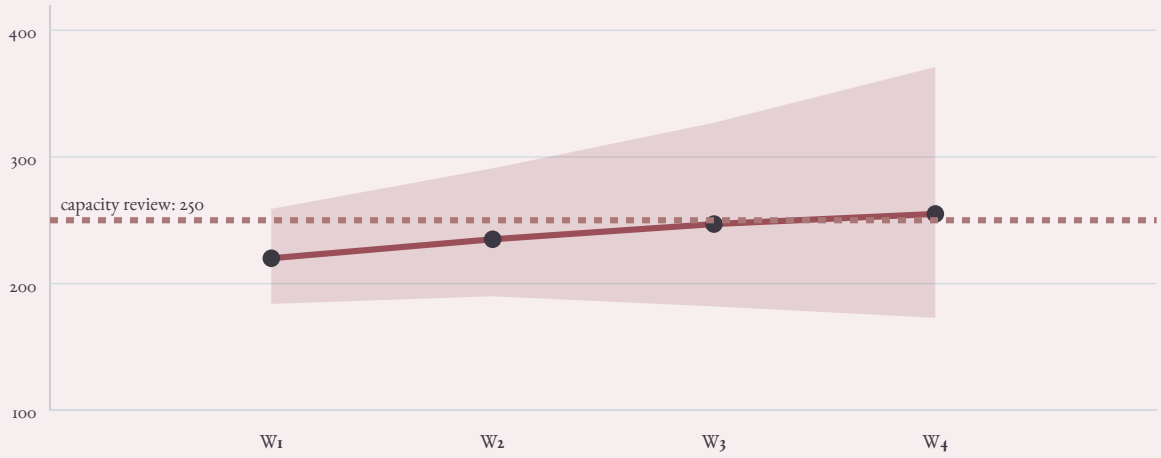
APPENDIX C / SAMPLE REPORT

Quantitative outlook

Forecast quantiles are illustrative values selected to demonstrate reporting style. They are not the result of a validated fitted model.

Horizon	10th percentile	Median	90th percentile	Persistence
Week 1	184	220	259	202
Week 2	190	235	291	202
Week 3	182	247	327	202
Week 4	173	255	371	202

FOUR-WEEK SYNTHETIC ADMISSIONS FORECAST



Shaded area: 80% central interval; all values fictitious

How to read the graphic

The line marks illustrative medians and the shaded band marks a central 80% interval. The threshold at 250 is an invented capacity-review level; it is not a universal health-policy trigger. The graphic alone does not identify the probability of exceeding the threshold because only selected quantiles have been provided.

APPENDIX C / SAMPLE REPORT

Source, model and uncertainty note

The format below is what a reviewer should require in a live briefing. For this fictional example, the sources are synthetic design placeholders rather than the official feeds cited in the reading list.

Source manifest

01	Target: fictional weekly facility-reported respiratory hospital admissions; last eligible value 202.
02	Context: synthetic emergency-visit trend, example laboratory-reporting notice and static district descriptors.
03	Provenance: demonstration manifest demo-manifest-01; no live external sources ingested.

Method and validation

A candidate multi-horizon Transformer is shown alongside persistence. The quantiles are illustrative, not calibrated by a real historical backtest. A live report would include model version, model-selection period, rolling-origin evaluation dates, WIS against baseline, interval coverage, failure rates and issue-time data quality. No evidence of actual predictive skill can be inferred from this exercise.

Decision language

Planners may wish to review contingency staffing and supply options if their own verified operational threshold is approached. The model does not make a clinical recommendation, establish a causal explanation or supersede local capacity data. New evidence should trigger a revised report with change notes.

APPENDIX C / SAMPLE REPORT

Review and publication record

The human review record protects the distinction between quantitative output and an authorized public-health statement. The following form illustrates sign-off fields.

Field	Review requirement
Issue identifier	Unique publication ID and UTC issue time
Definition check	Target, geography, units, horizon and truth version verified
Source health	Freshness, missingness and revision status accepted or qualified
Model check	Baseline, quantile ordering, evaluation validity and abstention rules
Claim tracing	Every qualitative statement linked to a source or labeled judgment
Numbers check	Narrative figures match forecast JSON exactly
Ethics	Aggregation, access rights, privacy and potential harms reviewed
Approval	Named analyst, supervisor where necessary, and archived edition

PUBLICATION RULE	If the target definition, source status, or numerical narrative fails review, fix the report or abstain. A polished layout cannot compensate for unresolved data problems.

APPENDIX D

The model comparison workbook

This workbook turns general advice into acceptance tests. Its sample figures are fabricated for instruction; they must never be presented as observed model performance.

D.1 Evaluation contract

Question	Required answer
What is the target?	Exact outcome, geography, time and unit
What is the origin?	Issue time, cut-off and accepted vintages
What is the history?	Train/validation/test periods and origin list
What is the comparator?	Persistence plus meaningful seasonal/statistical baseline
What is scored?	Median error, WIS, interval coverage, missed outputs
What gets stratified?	Horizon, district, phase, reporting completeness
What is locked?	Feature recipe, hyperparameters and final test protocol
What blocks release?	Leakage, nonmonotone quantiles, unavailable evidence

HARD RULE	An artifact cannot be considered forecast-validated if its retrospective features did not obey historical source availability.

APPENDIX D / CONTINUED

D.2 A deliberately imperfect score sheet

Suppose a fictional rolling-origin exercise yields these numbers. The values illustrate how to discuss a model that improves a single score while still failing a reliability criterion.

Illustrative metric	Persistence	Statistical	Transformer
WIS (lower is better)	49	45	43
80% interval coverage	0.79	0.77	0.69
4-week MAE	36	31	30
Missing forecast rate	0.0%	0.0%	1.5%

Interpretation

The Transformer has the lowest WIS in this invented table but its 80% interval captures only 69% of outcomes. That undercoverage is operationally important. A release decision could call for recalibration, investigation of forecast origins with rapid trends, or use of a more reliable ensemble. Reporting only the WIS ranking would conceal the limitation.

Next experiment

Compare calibration adjustment on a distinct validation interval, score it on untouched later origins and record whether improved coverage comes at the cost of excessive interval width. Investigate whether undercoverage concentrates around epidemic peaks or missing-data periods.

APPENDIX D / CONTINUED

D.3 A model failure taxonomy

Use a failure register during weekly analyst meetings. The purpose is to design targeted corrective actions rather than re-train indiscriminately.

Target or label failure

Definition changed; observations incomplete; truth revised later.

Availability failure

An input became available only after the simulated issue time.

Distribution failure

Intervals are too narrow, quantiles cross or counts turn negative.

Shift failure

Variant, intervention or reporting pattern differs from training regimes.

Aggregation failure

National total conflicts with subregional estimates.

Communication failure

Narrative invents probability, quote, reason or action.

Workflow failure

Run finishes late, missing sources are silently forward-filled, or publication bypasses approval.

REVIEW QUESTION	Which failure would cause the most harm in your application, and how would your pipeline detect it before release?

APPENDIX E

Glossary of analysis and forecasting terms

The definitions are concise reading aids. Their precise operational meaning depends on the chosen target, source and model.

Ablation Controlled removal of a component to assess its contribution.

Abstention Deliberate decision not to publish an unreliable forecast.

Analytic judgment Assessment that goes beyond directly observed facts.

Architecture ablation Controlled test of a modeling design component.

Artifact lineage Traceable relationship among inputs, models and outputs.

Availability time Earliest moment an observation could enter a forecast.

Calibration Agreement between predicted probabilities and observed frequencies.

Channel Individual variable within a multivariate series.

Common-mode failure Multiple systems failing because they share an assumption or dependency.

Context length Amount of historical data supplied to a forecast.

Coverage People, facilities, geography or events represented by a dataset.

Coverage bias Systematic gaps in represented populations or places.

Cross-attention Attention between representations of different token sets.

Data contract Explicit schema and semantics enforced between components.

Data minimization Collecting no more data than the declared purpose requires.

Data vintage Snapshot of what values were published at a given time.

APPENDIX E / GLOSSARY CONTINUED

Decision loss Cost of actions and misses under a stated policy.

Decision threshold User-defined value at which an action may be considered.

Decision-ready report Output connecting evidence, uncertainty and possible action.

Denominator Population or tested group relative to which an event is counted.

Drift Change in data or model performance over time.

Early stopping Ending training when held-out validation performance stops improving.

Ecological fallacy Mistaking group-level relationships for individual-level effects.

Effective date Date when a contextual value becomes applicable.

Embedding Vector encoding of text or other input.

Ensemble Combination of forecasts from multiple models.

Entity resolution Matching mentions to the same real-world person, place or event.

Event-based surveillance Detection and assessment of discrete potentially significant reports.

Evidence packet Selected, traceable source material supporting a report.

Evidence retrieval Finding passages that substantiate a report claim.

Experiment registry Versioned record of configurations, inputs, artifacts and scores.

Feature registry Contract describing how every model input is constructed.

APPENDIX E / GLOSSARY CONTINUED

Feedback effect Forecast-triggered action that changes later observations.

Forecast origin Last moment at which evidence is permitted into a forecast.

Forecast packet Versioned set of validated predictions and metadata.

Forecast skill Performance improvement relative to a named reference method.

Gating Learned mechanism that suppresses or controls transformed information.

Granularity Temporal and geographic resolution of a target.

Horizon Distance from issue time to the future target.

Idempotence Repeated ingestion does not duplicate or corrupt records.

Immutable raw zone Preserved source files that are not overwritten by transformations.

Incidence Number of new events during a stated period.

Indicator-based surveillance Routine measurement of predefined health indicators.

iTransformer Architecture representing series variables as Transformer tokens.

Key assessment Short statement of the most decision-relevant conclusion.

Known-future covariate Future input truly known when issuing the prediction.

Look-ahead bias Advantage from data unavailable at the historic forecast time.

Loss function Quantity minimized during model training.

APPENDIX E / GLOSSARY CONTINUED

MAE Mean absolute error of point forecasts.

Mask Restriction that prevents attention to invalid positions.

Mechanistic model Model built around assumed processes rather than associations alone.

Model disagreement Divergence among candidate predictions.

Multi-horizon Producing forecasts for several future time steps.

Multimodal fusion Combining distinct data modalities under explicit semantics.

Nowcast Estimate of a recent incompletely observed quantity.

Patch Contiguous segment of time steps treated as a learned token.

PatchTST Patch-based Transformer architecture for time-series forecasting.

Persistence Forecast repeating the most recent observation.

Pinball loss Asymmetric error score used for quantile predictions.

Postmortem Documented analysis of a failure and its corrective action.

Prediction interval Range estimated to contain a future realization with stated coverage.

Primary source Original publisher or recorder of an observation.

Provenance Recorded origin and transformation history of information.

Purpose limitation Restricting reuse to justified and authorized aims.

APPENDIX E / GLOSSARY CONTINUED

Quantile Value below which a specified fraction of a distribution falls.

Quantile crossing Predicted higher quantile falling below a lower quantile.

Query-key-value Learned projections used to compute attention.

RAG Retrieval-augmented generation grounded in selected documents.

Regime shift Change in the mechanism generating observed data.

Release gate Automated or human condition required before publication.

Reporting lag Delay from underlying event to data publication.

RMSE Root mean squared point error, emphasizing large misses.

Robustness Ability to maintain acceptable behavior under perturbation or shift.

Rolling origin Chronological validation with advancing issue dates.

Schema drift Change in field names, meanings or allowed values over time.

Seasonal naive Forecast repeating an analogous seasonal observation.

SEIR Compartmental model separating susceptible, exposed, infectious and removed groups.

Shadow deployment Running a model without using its outputs for live decisions.

Softmax Normalization mapping scores into positive weights summing to one.

Source portfolio Selected set of feeds serving defined analytical roles.

APPENDIX E / GLOSSARY CONTINUED

Static covariate Attribute treated as constant across the forecast window.

Static encoder Component incorporating time-invariant contextual features.

Stride Step size between successive sequence patches.

Syndication Redistribution of one source through other publishers.

Synthetic case Educational illustration using invented rather than observed outcomes.

Target Precisely defined quantity a model is scored against.

Temporal leakage Future information entering a past model input.

Temporal token Representation associated with one observation time.

Tensor Multidimensional array of numerical values.

TFT Temporal Fusion Transformer for mixed-covariate multi-horizon forecasting.

Time series Measurements indexed by ordered observation periods.

Transportability Ability to use learned relations in a different context.

Triangulation Comparison across genuinely independent evidence streams.

Truth vintage Version of realized target values selected for evaluation.

WIS Proper scoring rule for collections of predictive intervals and medians.

APPENDIX F / ADVANCED METHODS

F.1 Attention: the computation behind the story

A Transformer is not an agent reasoning over weeks in natural language. It transforms numeric tensors. For a single attention head, Q contains query vectors, K key vectors and V value vectors. If the key dimension is d_k, the scaled dot-product attention output is:

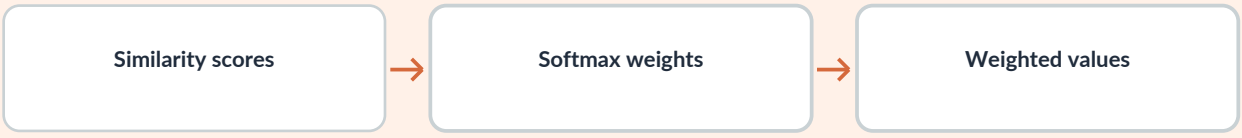
```
Attention(Q,K,V) = softmax(Q @ K.T / sqrt(d_k)) @ V
```

The softmax operates across eligible key positions. A mask can suppress invalid or unavailable positions. Multiple heads form different learned projections and their outputs are recombined. The familiar formula does not encode epidemiological causality; it describes how representations are mixed.

A numerical micro-example

Assume one query produces already scaled similarity scores 0.60, 0.05 and -0.20 across three past positions. Their exponentials are approximately 1.82, 1.05 and 0.82, so the softmax weights are about 0.494, 0.285 and 0.222. If the corresponding one-dimensional values are 2, 5 and 10, the weighted representation is about 4.63. This is a feature computation, not a probability of disease spread.

CONCEPT MAP



Expert interpretation

In a time-token encoder, the dot products compare temporal representations. In an inverted architecture, they may compare variable representations. Position information matters because sequence order has a physical meaning. Attention heads and outputs can be unstable under correlated features or equivalent reparameterizations, so they should not be interpreted as calibrated causal attributions. The proper question is whether the resulting forecasts improve in held-out settings.

APPENDIX F / ADVANCED METHODS

F.2 Covariate roles in multi-horizon learning

At issue time t, an H-step model may consume historic observed variables x[t-L+1:t], known future calendar variables k[t+1:t+H], and static regional metadata s. It must not consume future values of x unless those values really are known when forecasting. This is the practical motivation for the covariate separation in Temporal Fusion Transformer.

Input role	Example	Status at issue time
Observed historical	Admissions and test positivity	Only past releases
Known future	Week number, scheduled holiday	Available for all H steps
Static context	Population, region category	Effective-date snapshot
Scenario condition	Hypothetical future policy	Assumption, not observation

A useful loss design

For each horizon h, the model may produce q(0.1,h), q(0.5,h) and q(0.9,h). Enforce q(0.1,h) <= q(0.5,h) <= q(0.9,h), and for an admission count require nonnegative values. Estimate a central 80% interval from the outer quantiles. Test coverage by horizon and by local epidemic phase: wide national coverage can conceal undercoverage in a small district at a surge peak.

The gating and variable-selection components of TFT produce diagnostics, not a causal decomposition. Correlated ED visits and test positivity may exchange their apparent importance across model refits. A strong publication workflow treats those diagnostics as prompts for further investigation, not definitive reasons for a forecast.

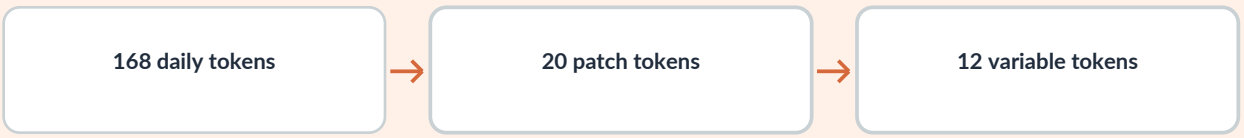
EVALUATION RULE	Give every architecture identical issue-time data and compare it on the same pre-specified rolling origins.

APPENDIX F / ADVANCED METHODS

F.3 Tokenization changes what attention means

Suppose a daily input window contains 168 observations and 12 measured channels. A conventional time-token Transformer can form 168 tokens, each representing one day across channels. A patch-based design might form overlapping segments of length P=16 with stride S=8. That produces floor((168-16)/8)+1 = 20 patch tokens per channel, before any architecture-specific padding. An inverted design may instead begin with 12 variable tokens, each encoding an entire historical channel.

CONCEPT MAP



Design	Main attention relation	Risk to test
Time tokens	Relationships across days	Cost on long histories
PatchTST-style	Relationships across temporal patches	Abrupt spikes hidden by patching
iTransformer-style	Relationships across measured variables	Missing or changing feature roster

Ordinary full self-attention forms a quadratic-size attention matrix in the number of tokens. Reducing the token count from 168 to 20 greatly reduces that particular matrix, but does not imply the whole model becomes equally faster: patch embedding, channel repetition and other layers also consume resources.

Evaluation question

Do six, twelve or twenty-four months of history improve a real four-week admission forecast when source definitions changed during the period? The answer must come from matched backtests. Longer context is neither intrinsically more informative nor intrinsically safer.

APPENDIX F / ADVANCED METHODS

F.4 Quantile loss and weighted interval score

For an observed outcome y and predicted quantile q at probability level a, define error e=y-q. Pinball loss is a*e if e is nonnegative, and (a-1)*e otherwise. The asymmetry causes a quantile to estimate the appropriate side of a distribution rather than the mean. Averaging the loss across several target horizons and locations is a common training objective, but the evaluation should still be separated by horizon.

```
rho_a(e) = max(a * e, (a - 1) * e)

Example at y = 300:
  q10 = 200  -> rho_0.1 = 10
  q50 = 250  -> rho_0.5 = 25
  q90 = 330  -> rho_0.9 =  3
```

An interval score evaluates a central interval [l,u] with nominal miscoverage a. It rewards narrow intervals but adds a penalty when the realized outcome falls below l or above u. Weighted interval score combines a median absolute-error term and a collection of interval scores; see Bracher et al. (2021) for the formal scoring definition and its properties.

```
IS_a(l,u;y) = (u-l)
             + 2/a * (l-y) if y < l
             + 2/a * (y-u) if y > u
(only the applicable penalty term is added)
```

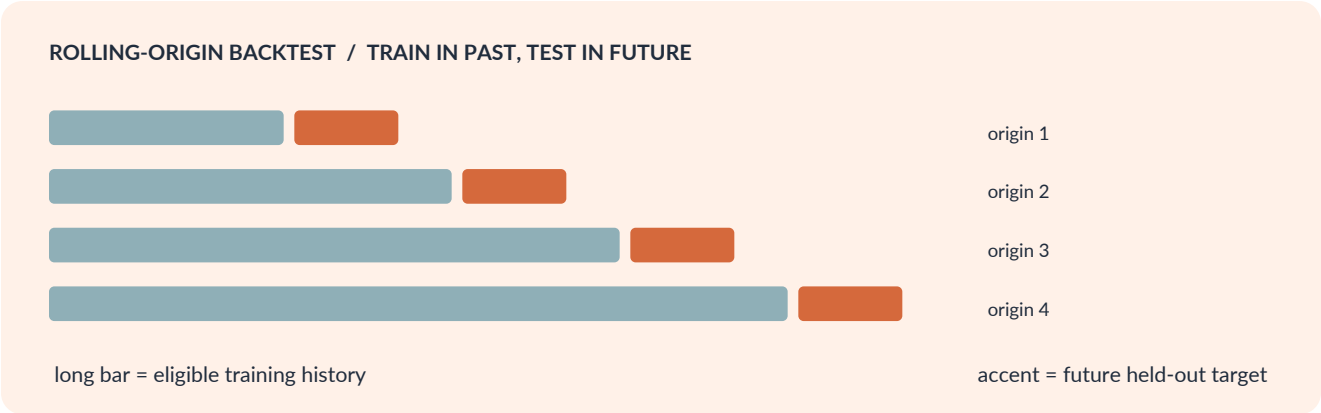
Calibration is not accuracy in disguise

Across many held-out forecasts, an interval labeled 80% should contain roughly 80% of corresponding future outcomes under a well-calibrated stable setting. An interval can achieve coverage by becoming uselessly wide; that is why a proper score and interval width matter. Coverage guarantees from some postprocessing approaches may fail under strong distribution shift or dependent observations.

APPENDIX F / ADVANCED METHODS

F.5 Vintage-aware rolling-origin backtesting

The hardest leakage bugs are often not obvious future rows. They arise when an old observation is revised later and historical training silently reads the new value. A robust backtest preserves, or reconstructs, what was actually released at each past origin. The target truth used for scoring may be finalized later under a written policy; the model input for that origin may not.




```
for issue_time in predefined_origins:
    raw = read_archived_sources(as_of=issue_time)
    features = build_features(raw, cut_off=issue_time)
    assert features.max_release_time <= issue_time
    model = fit_on_prior_allowed_labels(issue_time)
    pred = model.predict_quantiles(features)
    archive(issue_time, raw.manifest, pred)

for record in archived_predictions:
    truth = get_final_truth(record.target_period)
    score(record.prediction, truth)
```

The two loops embody a crucial separation: issue-time simulation and later scoring. Do not regenerate past predictions with current source tables and call it a historical evaluation. If archived vintages do not exist, disclose the limitation and conduct a forward prospective shadow period before operational claims.

AUDIT ARTIFACT	Every scored forecast should have an issue time, input manifest, model version, output quantiles and final truth definition.

APPENDIX F / ADVANCED METHODS

F.6 Text and numeric data: one pipeline, two truth tests

A numeric indicator can be validated against a counting rule. A news or public-document event is validated through attribution, original passages, independence, geographic identity and publication timing. A multimodal pipeline should preserve both kinds of evidence while making the quantitative model robust to media attention and duplicate coverage.

An event object with the right clocks

```
{
  "type": "laboratory_reporting_change",
  "location_id": "fictional-district-01",
  "publication_time": "2026-10-07T10:00:00Z",
  "effective_date": "2026-10-06",
  "source_id": "synthetic-notice-01",
  "verified": true,
  "source_span": "... synthetic quoted excerpt ..."
}
```

The effective date may precede the publication date. It still cannot become an issue-time feature before the publication exists. If a large language model extracts the event, preserve the quoted span, capture the extractor version and evaluate extraction errors including negation, uncertainty and multilingual place names. Use retrieval to cite that source later; do not ask a text generator to infer an unprovided numerical forecast.

A useful fusion ablation

Test three models on the same held-out origins: numeric surveillance only, numeric plus verified event flags, and numeric plus learned text embeddings. Include a news-volume control. Improvements should survive across regions and across periods of unequal media interest. A model that improves only during heavy news coverage may be exploiting publication volume rather than disease dynamics.

APPENDIX F / ADVANCED METHODS

F.7 From a threshold probability to a decision

A threshold event, such as weekly admissions exceeding a user-defined planning level, converts a full distribution into a probability. But probability is not a recommendation until the consequences of actions and misses are specified. Even a well-calibrated model does not know the cost of calling in additional staff or the harm of insufficient capacity.

A deliberately simplified decision rule

Let p be the calibrated probability that demand exceeds the defined threshold. Assume acting unnecessarily costs C_FA, and not acting when demand does exceed the threshold costs C_MISS. If the action entirely avoids that miss cost and has no other effects, the expected losses are C_FA*(1-p) for acting and C_MISS*p for not acting. Under these restrictive assumptions, choose the action only when p > C_FA/(C_FA+C_MISS).

```
P(event) = p
Loss(act)    = C_FA   * (1 - p)
Loss(no act) = C_MISS * p
Threshold    = C_FA / (C_FA + C_MISS)
```

Example: if a false alarm costs 1 unit and a miss costs 4 units, the threshold is 1/(1+4)=0.20. That is a mathematical teaching example, not a public-health policy. Real intervention effects, resource constraints and ethical considerations make the utility model more complicated.

CRITICAL LIMIT	Do not compute p from a median and an 80% interval alone. Use a valid predictive distribution or forecast samples; then seek user-approved decision costs.

APPENDIX F / ADVANCED METHODS

F.8 Contracts between the Transformer and the report writer

A generative language model may be built on Transformer technology, but it should not be confused with a statistical Transformer trained to forecast admissions. The two systems have different objective functions and failure modes. Separate them with machine-checkable interfaces.

CONCEPT MAP



The numeric model should emit:

01 A complete list of horizon-specific quantiles or samples, target metadata, issue time, feature snapshot, model identifier and calibrated score summary.

02 Explicit quality flags for missingness, data recency, revisions and whether the release is fully supported or conditional.

03 The reference model forecast and registered validation evidence used to compare any claim of improvement.

The report writer should emit:

01 Observed facts with source IDs, predicted outcomes with forecast IDs, and clearly labeled scenarios with assumptions.

02 A figure-and-prose consistency check that rejects any number not in the signed forecast packet.

03 A human review and publication record, including limitations, revision history and triggers for a new assessment.

The final intelligence report is a controlled communication product. In this design, the language model can improve readability while statistical validation remains in the forecasting service and factual traceability remains in the evidence store.

REFERENCE LIBRARY

Annotated sources and primary research

Use original technical and public-health sources. These references support the methods discussed in the book; they do not supply the fictional case numbers.

[01] **World Health Organization —** Epidemic Intelligence from Open Sources (EIOS), current initiative and methods portal
Official foundation for event-based public-health intelligence and collaborative use of open data. Open the source

[02] **World Health Organization —** EIOS Strategy 2024-2026 (published 11 September 2025)
Illustrates institutional governance, collaboration and the limits of platform-first solutions. Open the source

[03] **World Health Organization —** Defining collaborative surveillance (2024)
Framework for integrated multi-source surveillance across organizations and sectors. Open the source

[04] **US Centers for Disease Control and Prevention —** Respiratory Virus Hospitalization Surveillance Network (RESP-NET), dashboard (2026)
Example of defined hospitalization outcomes; verify each geography and downloadable series before use. Open the source

[05] **US Centers for Disease Control and Prevention —** FluSight 2025-2026 evaluation (30 September 2026)
Operational forecasting benchmark; warns of reduced skill around rapidly changing trends. Open the source

REFERENCE LIBRARY / CONTINUED

[06] **World Health Organization —** Guidelines on ethical issues in public health surveillance (2017)
Foundational discussion of privacy, proportionality, fairness and oversight. Open the source

[07] **Hyndman & Athanasopoulos —** Forecasting: Principles and Practice, third edition: time-series cross-validation
Accessible explanation of rolling-origin validation and multi-step forecasting. Open the source

[08] **Lim, Arik, Loeff & Pfister —** Temporal Fusion Transformers for interpretable multi-horizon time series forecasting (IJF, 2021)
Primary paper for static, observed historical and known-future covariates in TFT. Open the source

[09] **Nie et al. —** A Time Series is Worth 64 Words: Long-term Forecasting with Transformers (ICLR, 2023)
Primary paper for PatchTST tokenization and channel independence. Open the source

[10] **Liu et al. —** iTransformer: Inverted Transformers Are Effective for Time Series Forecasting (ICLR, 2024)
Primary paper for variable-token attention and alternative series representations. Open the source

REFERENCE LIBRARY / CONTINUED

[11] **Bracher, Ray, Gneiting & Reich —** Evaluating epidemic forecasts in an interval format (PLOS Comp Biol, 2021)
Introduces practical evaluation and interpretation of weighted interval score. Open the source

[12] **Vaswani et al. —** Attention Is All You Need (NeurIPS, 2017)
Introduces the Transformer attention architecture and foundational notation. Open the source

[13] **Collins et al. —** TRIPOD+AI reporting guideline (BMJ, 2024)
Reporting discipline for clinical prediction model studies; useful but not a substitute for public-health oversight. Open the source

[14] **NIST —** Artificial Intelligence Risk Management Framework 1.0 (2023)
Framework for trustworthy AI design, testing and governance. Open the source

[15] **WHO —** Global Health Observatory data portal
Global indicators and documentation; dataset availability differs by indicator and channel. Open the source

ACCESS NOTE	Portal names and download procedures can change. Verify last update, file definitions, access rights, coverage and any revisions before deploying a live ingestion connector.

CLOSING NOTE

The forecast is an accountable argument

The professional standard for a prediction intelligence report is a traceable chain: a well-defined decision, an honest representation of observations, a disciplined forecast, an evaluation that respects past information, and language that makes uncertainty actionable. Transformers belong in this chain as tested models. They are not replacements for judgment, independent evidence or public-health expertise.

A reader who can reproduce the input vintage, recreate the predicted distribution, check its coverage and explain the final report has achieved the central objective of this book.